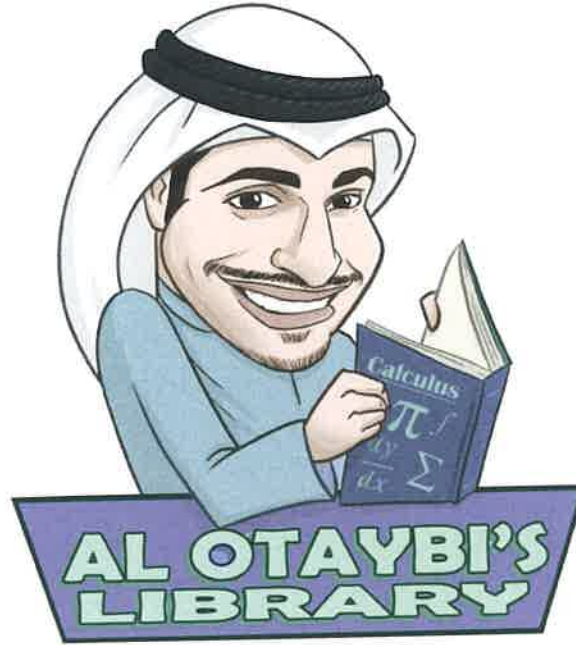


Kuwait University

2020/2021

Fall course



Calculus A

Second

مراجعة

اختبارات سابقة

(مراجعة السكند)

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Evaluate the following limit if it exist

$$\lim_{x \rightarrow \pi} \frac{\tan x}{x - \pi}$$

$\frac{0}{0}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f(x) = \tan x$$

$$a = \pi$$

$$f'(x) = \sec^2 x$$

$$f'(a) = f'(\pi) = \sec^2 \pi = \frac{1}{\cos^2 \pi} = \frac{1}{(-1)^2} = 1$$

$$\therefore \lim_{x \rightarrow \pi} \frac{\tan x}{x - \pi} = 1$$

$$\lim_{x \rightarrow 3} \frac{\ln x - \ln 3}{x - 3}$$

$\frac{0}{0}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f(x) = \ln x$$

$$a = 3$$

$$f'(x) = \frac{1}{x}$$

$$f'(a) = f'(3) = \frac{1}{3}$$

$$\therefore \lim_{x \rightarrow 3} \frac{\ln x - \ln 3}{x - 3} = \frac{1}{3}$$

$$\lim_{x \rightarrow 1} \frac{e^{x^2 - x} - 1}{x - 1}$$

$\frac{e^0 - 1}{0} = \frac{0}{0}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f(x) = e^{x^2 - x}$$

$$a = 1$$

$$f'(x) = e^{x^2 - x} (2x - 1)$$

$$f'(a) = f'(1) = e^0 (2 - 1) = e^0 (1) = 1$$

$$\therefore \lim_{x \rightarrow 1} \frac{e^{x^2 - x} - 1}{x - 1} = 1$$

$$\lim_{x \rightarrow 0} \frac{3^x - 1}{x}$$

$\frac{0}{0}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f(x) = 3^x$$

$$a = 0$$

$$f'(x) = 3^x (1) \ln 3$$

$$f'(a) = f'(0) = 3^0 \ln 3 = \ln 3$$

$$\therefore \lim_{x \rightarrow 0} \frac{3^x - 1}{x} = \ln 3$$

Evaluate the following limit if it exist

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2 + x - 6}$$

$\frac{0}{0}$

Sol #1

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)(x+3)}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{(x-2)} \frac{\sin(x-2)}{x-2}}{\cancel{(x-2)}(x+3)}$$

$$= \frac{1}{5}$$

Sol #2

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)(x+3)}$$

$$= (1) \frac{1}{5} = \frac{1}{5}$$

$$\lim_{x \rightarrow 2} \frac{\cos(x-2) - 1}{x^2 + x - 6}$$

$\frac{1-1}{0} = \frac{0}{0}$

$$= \lim_{x \rightarrow 2} \frac{\cos(x-2) - 1}{(x-2)(x+3)}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{(x-2)} \frac{\cos(x-2) - 1}{x-2}}{\cancel{(x-2)}(x+3)}$$

$$= \frac{0}{5} = 0$$

Evaluate the following limit if it exist

$$\lim_{x \rightarrow \infty} \frac{\sinh x}{e^x}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{2e^x}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{2e^x} \quad \frac{\infty - 0}{\infty} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x \left[1 - \frac{e^{-x}}{e^x} \right]}{2e^x}$$

$$= \frac{1 - 0}{2} = \frac{1}{2}$$

$$\lim_{x \rightarrow \infty} \coth x$$

$$= \lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x \left[1 + \frac{e^{-x}}{e^x} \right]}{e^x \left[1 - \frac{e^{-x}}{e^x} \right]}$$

$$= \frac{1 + 0}{1 - 0} = 1$$

Kuwait University
Department of Mathematics

Calculus I (Math 101)
Second Exam

16 April 2016
Duration 90 minutes

Answer all questions. Calculators and mobile phones are not allowed.

Q1. [10 pts.] Evaluate the following limit, if it exists:

$$\lim_{x \rightarrow 2} \frac{2x^2 - x - 6}{\sin(x - 2)}$$

Q2. [10 pts.] The surface area of a sphere increases at the rate of $48 \pi \text{ cm}^2/\text{sec}$. Find the rate of change of the volume when the surface area is 36 cm^2 .

Q3. [5+5=10 pts.] Let $f(x) = \ln x$.

- a) Find the linearization of f at $a = 1$.
- b) Use linear approximation to estimate $\ln(1.01)$.

Q4. [10 pts.] Show that, if $a > 0$, then the equation: $x^3 + ax - 1 = 0$ has at most one real root.

Q5. [5+5+10=20 pts.]

a) Show that $\frac{d}{dx} (\sin^{-1}(2x)) = \frac{2}{\sqrt{1-4x^2}}$

b) Find $\frac{dy}{dx}$ if: $y = \frac{x}{\sin^{-1}(2x)}$

c) Find $\frac{dy}{dx}$ if: $y = (3x^2 - 2x + 1)^{\tan x}$.

Q6. [40 pts.] Let $f(x) = \frac{x^2 + x - 1}{x - 1}$.

a) Find the vertical and horizontal asymptotes for the graph of f , if any.

b) Given that $f'(x) = \frac{x^2 - 2x}{(x - 1)^2}$.

- i) Find the intervals on which f is increasing and the intervals on which f is decreasing.
- ii) Find the local maximum and minimum values of f , if any.

c) Given that $f''(x) = \frac{2}{(x - 1)^3}$.

- i) Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward.
- ii) Find the points of inflection, if any.

d) Sketch the graph of f .

Q#1

$$\lim_{x \rightarrow 2} \frac{2x^2 - x - 6}{\sin(x - 2)} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 2} \frac{(2x + 3)(x - 2)}{\sin(x - 2)}$$

Solution #1

$$= \lim_{x \rightarrow 2} \frac{(2x + 3)(x - 2)}{(x - 2) \frac{\sin(x - 2)}{x - 2}}$$

$$= \frac{4 + 3}{(1)}$$

$$= \frac{7}{1} = 7$$

Solution #2

$$= (4 + 3) (1)$$

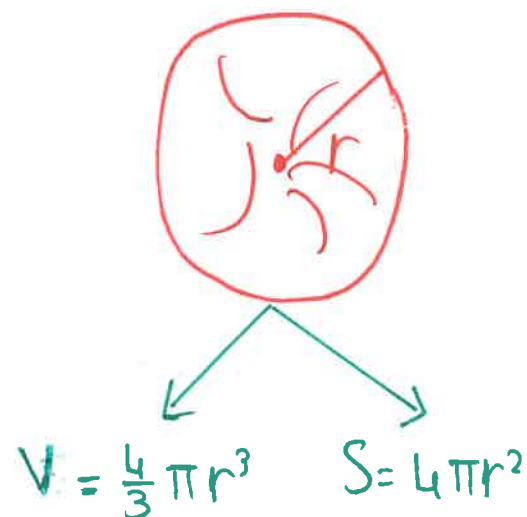
$$= 7$$

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Q#2

$$\frac{dS}{dt} = 48\pi \text{ cm}^2/\text{sec}$$

$$\frac{dV}{dt} ?? \rightarrow S = 36 \text{ cm}^2$$



$$V = \frac{4}{3} \pi r^3$$

Diff. w.r.t. t

$$(1) \frac{dV}{dt} = \frac{4}{3} \pi \cancel{3} r^2 \frac{dr}{dt}$$

$$= 4\pi \left(\frac{3}{\sqrt{\pi}} \right)^2 (2\sqrt{\pi})$$

$$= 4\pi \frac{9}{\pi} 2\sqrt{\pi}$$

$$= 72\sqrt{\pi} \text{ cm}^3/\text{sec.}$$

$$S = 4\pi r^2$$

$$36 = 4\pi r^2$$

$$\div 4\pi$$

$$r^2 = \frac{36}{4\pi}$$

$$r^2 = \frac{9}{\pi} \rightarrow r = \frac{3}{\sqrt{\pi}} \text{ cm}$$

$$S = 4\pi r^2$$

Diff. w.r.t. t

$$\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

$$48\pi = 8\pi \frac{3}{\sqrt{\pi}} \frac{dr}{dt}$$

$$48\pi = 24\sqrt{\pi} \sqrt{\pi} \frac{1}{\sqrt{\pi}} \frac{dr}{dt}$$

$$\div 24\sqrt{\pi}$$

$$\frac{dr}{dt} = \frac{48\pi}{24\sqrt{\pi}} \rightarrow \frac{dr}{dt} =$$

$$\frac{dr}{dt} = 2\sqrt{\pi} \text{ cm/sec}$$

Q#3

Let $f(x) = \ln x$

a) Find linearization at $a=1$

b) Use Linear approximation to estimate $\ln(1.01)$

a) $f(x) = \ln x$

$$x_0 = 1$$

$$x_1 = x \longrightarrow \Delta x = x - 1$$

$$f(x_0) = \ln 1 = 0$$

$$f'(x) = \frac{1}{x}$$

$$f'(x_0) = \frac{1}{1} = 1$$

$$\begin{aligned} L(x) &= f(x_0) + f'(x_0) \Delta x \\ &= 0 + (1)(x-1) \\ &= x-1 \end{aligned}$$

$$\begin{aligned} b) \ln(1.01) &\approx 1.01 - 1 \\ &\approx 0.01 \end{aligned}$$

Q #4

Show that if $a > 0$ then
 $x^3 + ax - 1 = 0$ has at most one real root.

Let $f(x) = x^3 + ax - 1$

Let x_1, x_2 are two different solutions of f .

1) Domain of $f(x) = \mathbb{R}$ [polynomial of $d=3$]
 $\therefore f(x)$ is continuous on $[x_1, x_2]$

2) $f'(x) = 3x^2 + a$
 $f(x)$ is differentiable on $\mathbb{R}$ [polynomial of $d=2$]
 $\therefore f(x)$ is differentiable on (x_1, x_2)

3) $f(x_1) = 0$
 $f(x_2) = 0 \longrightarrow \therefore f(x_1) = f(x_2)$

From ①②③ $\longrightarrow \therefore$ Rolle's Theorem is satisfied
then there is a number $c \in (x_1, x_2)$ such that

$f'(c) = 0$

$3c^2 + a = 0 \longrightarrow 3c^2 = -a$ (which is Wrong)
 $3c^2 > 0 \neq -ve$

$\therefore f(x)$ has at most one solution.

Q#5

a) Show that $\frac{d}{dx}(\sin^{-1}(2x)) = \frac{2}{\sqrt{1-4x^2}}$

① $y = \sin^{-1}(2x)$

② take sin for both sides

$$\sin y = \sin \sin^{-1}(2x)$$

$$\sin y = 2x$$

③ Diff. w.r.t. x

$$\cos y \cdot y' = 2$$

$$y' = \frac{2}{\cos y}$$

$$\cos^2 y + \sin^2 y = 1$$

$$\cos^2 y = 1 - \sin^2 y$$

$$y' = \frac{2}{\sqrt{1 - \sin^2 y}}$$

$$\cos y = \sqrt{1 - \sin^2 y}$$

$$y' = \frac{2}{\sqrt{1 - (2x)^2}}$$

$$\therefore y' = \frac{2}{\sqrt{1 - 4x^2}}$$

b) Find $\frac{dy}{dx}$ if $y = \frac{x}{\sin^{-1}(2x)}$

$$y' = \frac{\sin^{-1}(2x) (1) - x \cdot \frac{2}{\sqrt{1-(2x)^2}}}{(\sin^{-1}(2x))^2}$$

c) Find $\frac{dy}{dx}$ if $y = (3x^2 - 2x + 1)^{\tan x}$

$$\ln y = \ln (3x^2 - 2x + 1)^{\tan x}$$

$$\ln y = \tan x \ln (3x^2 - 2x + 1)$$

Diff. w.r.t. x

$$\frac{y'}{y} = \sec^2 x \ln(3x^2 - 2x + 1) + \tan x \cdot \frac{6x - 2}{3x^2 - 2x + 1}$$

$$\begin{aligned} y' &= y \left[\sec^2 x \ln(3x^2 - 2x + 1) + \tan x \frac{6x - 2}{3x^2 - 2x + 1} \right] \\ &= (3x^2 - 2x + 1)^{\tan x} \left[\sec^2 x \ln(3x^2 - 2x + 1) + \tan x \frac{6x - 2}{3x^2 - 2x + 1} \right] \end{aligned}$$

Q#6

$$f(x) = \frac{x^2 + x - 1}{x - 1}$$

1) Domain of $f(x)$: $x-1=0$
 $x=1$
 $\Rightarrow \therefore D_f = \mathbb{R} \setminus \{1\}$

2) Horizontal Asymptote

$$\lim_{x \rightarrow \pm\infty} \frac{x^2 + x - 1}{x - 1}$$
$$= \lim_{x \rightarrow \pm\infty} \frac{x^2 \left[1 + \frac{1}{x} - \frac{1}{x^2} \right]}{x \left[1 - \frac{1}{x} \right]} = \frac{\pm\infty [1+0-0]}{1-0} = \pm\infty$$

$\Rightarrow \therefore$ No H.A

3) Vertical Asymptote

$x=1$

$$\lim_{x \rightarrow 1^+} \frac{x^2 + x - 1}{x - 1} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 1^-} \frac{x^2 + x - 1}{x - 1} = \frac{1}{0^-} = -\infty$$

$\Rightarrow \therefore$ $x=1$ is V.A

$$f'(x) = \frac{x^2 - 2x}{(x-1)^2}$$

← → undefined (DNE)

$$= 0$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x=0 \quad x=2$$

∈ Domain ∈ Domain

$$x-1=0$$

$$x=1$$

∉ Domain



$f'(x)$	$(-\infty, 0)$	$(0, 1)$	$(1, 2)$	$(2, \infty)$
$x^2 - 2x$	+	-	-	+
$(x-1)^2$	+	+	+	+
	⊕	⊖	⊖	⊕

* interval of increasing

* interval of decreasing

at $x=0 \rightarrow$ L.max

$$y = \frac{0+0-1}{0-1} = 1$$

$\Rightarrow (0, 1)$ L.max

$(-\infty, 0)$ $(2, \infty)$

$(0, 1)$ $(1, 2)$

at $x=2 \rightarrow$ L.min

$$y = \frac{4+2-1}{2-1} = 5 = 5$$

$\Rightarrow (2, 5)$ L.min.

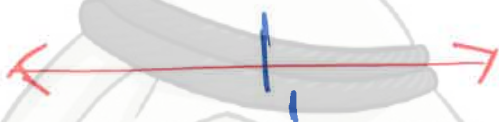
$$f''(x) = \frac{2}{(x-1)^3}$$

$= 0$
 $2 = 0$
 None

undefined (DNE)

$$x - 1 = 0$$

$x = 1 \notin \text{Domain}$



	$(-\infty, 1)$	$(1, \infty)$
$\frac{2}{(x-1)^3}$	+	+
	-	+
	⊖	⊕
	∩	∪

∴ $f(x)$ is concave down on $(-\infty, 1)$

∴ $f(x)$ is concave up on $(1, \infty)$

No inflection points

Key points

intersection point with H.A

None

$$\begin{array}{c|c} -1 & 3 \\ \hline & 1 \end{array}$$

$$\text{at } x = -1 \rightarrow y = \frac{1-1-1}{-1-1} = \frac{-1}{-2} = \frac{1}{2}$$

$$\text{at } x = 3 \rightarrow y = \frac{9+3-1}{3-1} = \frac{11}{2}$$

$$\Rightarrow \left(1, \frac{1}{2}\right) \quad \left(3, \frac{11}{2}\right)$$

* Intersection *

X-intercept

Y-intercept

$$0 = \frac{x^2 + x - 1}{x - 1}$$

* (x-1)

$$0 = x^2 + x - 1$$

$$a=1 \quad b=1 \quad c=-1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-1 \pm \sqrt{1 - 4(1)(-1)}}{2(1)}$$

$$= \frac{-1 \pm \sqrt{5}}{2}$$

$$\therefore P\left(\frac{-1 + \sqrt{5}}{2}, 0\right), P\left(\frac{-1 - \sqrt{5}}{2}, 0\right)$$

$$y = \frac{0+0-1}{0-1} = 1$$

$$(0, 1)$$

$$D_f = \mathbb{R} \setminus \{1\}$$

H.A $\rightarrow$ None

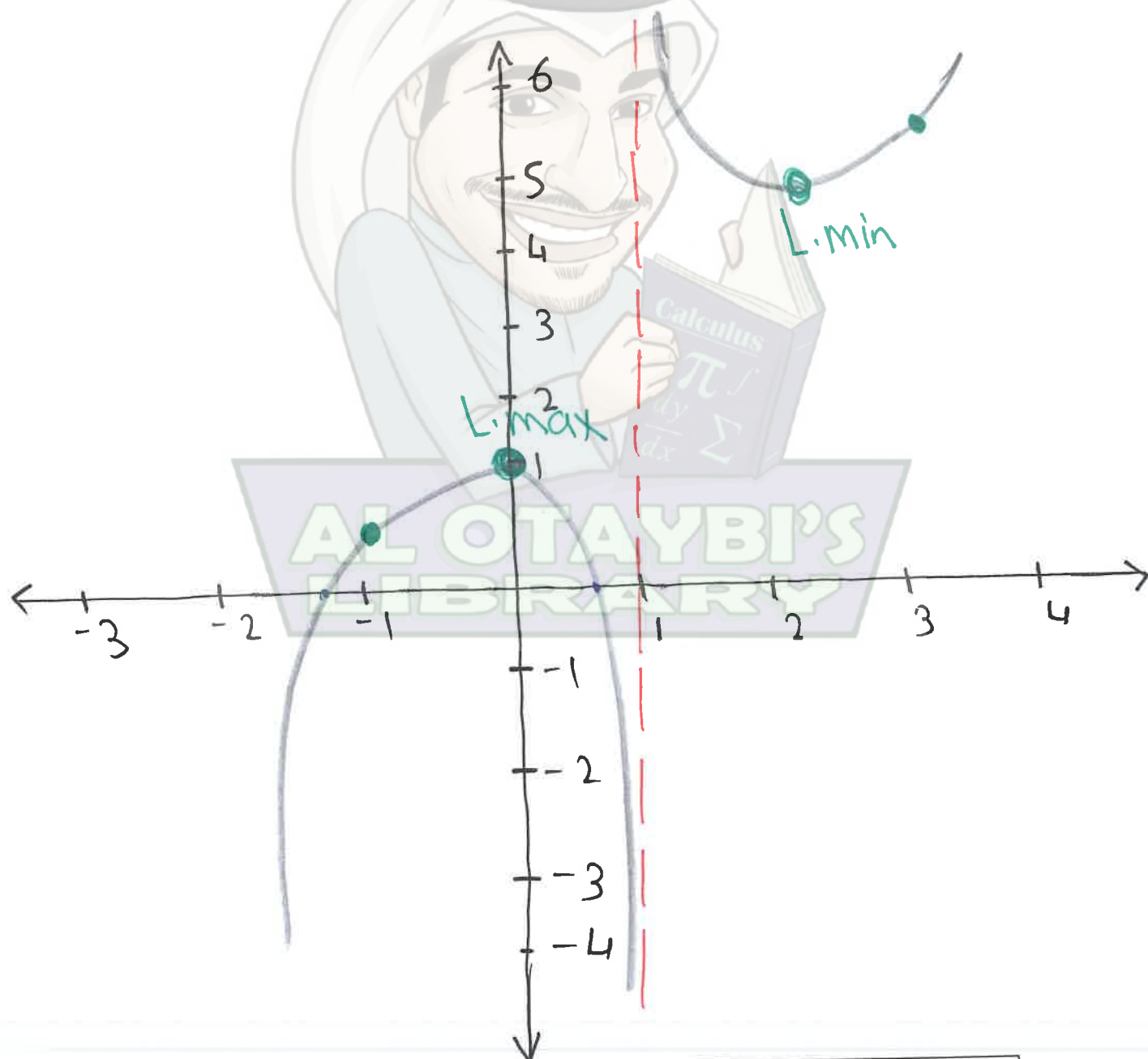
V.A $\rightarrow X=1 \begin{cases} \oplus & +\infty \\ \ominus & -\infty \end{cases}$

$(0,1)$ L.max , $(2,5)$ L.min

$\cup$ $(1,\infty)$

$\cap$ $(-\infty,1)$

$(-1, -\frac{1}{2})$ $(3, \frac{11}{2})$



Kuwait University
Department of Mathematics

Calculus I (Math 101)
Second Exam

18 April 2015
Duration 90 minutes

Answer all questions. Calculators and mobile phones are not allowed.

Q1. [10 pts.] Evaluate each of the following limits, if it exists:

(a) $\lim_{x \rightarrow 1} \frac{\sin^2(x-1)}{3(x^2 - 2x + 1)}$

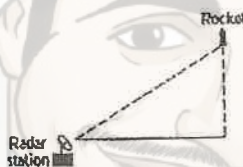
(b) $\lim_{x \rightarrow 0} \frac{3^x - 1}{x}$

Q2. [10 pts.] Find $\frac{dy}{dx}$, where $y = (\sin x)^{\ln x}$.

Q3. [10 pts.] Find an equation of the tangent line at the point $(0, 1)$ to the curve with equation

$$2y + x^2y + x \sin(x^2) = e^{xy} + 1.$$

Q4. [10 pts.] A rocket, rising vertically, is tracked by a radar station that is on the ground 4 km from the launchpad. How fast is the rocket rising when it is 3 km high and its distance from the radar station is increasing at a rate of 1500 km/h?



Q5. [10 pts.] Let $f(x) = e^x$.

(a) Find the linearization of f at $a = 0$.

(b) Use linear approximation to estimate $e^{0.01}$.

Q6. [10 pts.] Use the Mean Value Theorem to show that for all $x > 0$,

$$e^x > 1 + x.$$

Q7. [40 pts.] Let $f(x) = \frac{x^2 + 2x + 1}{x^2 - 4x + 4}$.

a) Find the vertical and horizontal asymptotes for the graph of f , if any.

b) Given that $f'(x) = -\frac{6(x+1)}{(x-2)^3}$.

i) Find the intervals on which f is increasing and the intervals on which f is decreasing.

ii) Find the local maximum and minimum values of f , if any.

c) Given that $f''(x) = \frac{6(2x+5)}{(x-2)^4}$.

i) Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward.

ii) Find the points of inflection, if any.

d) Sketch the graph of f .

Q#1

$$(a) \lim_{x \rightarrow 1} \frac{\sin^2(x-1)}{3(x^2-2x+1)} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{\sin^2(x-1)}{3(x-1)(x-1)}$$

$$= \lim_{x \rightarrow 1} \frac{\sin^2(x-1)}{3(x-1)^2}$$

Solution #1

$$= \lim_{x \rightarrow 1} \frac{(x-1)^2 \sin^2(x-1)}{3(x-1)^2}$$

$$= \frac{(1)}{3} = \frac{1}{3}$$

Solution #2

$$= \frac{1}{3} (1) = \frac{1}{3}$$

$$(b) \lim_{x \rightarrow 0} \frac{3^x - 1}{x} \quad \frac{3^0 - 1}{0} \quad \frac{1-1}{0} \quad \frac{0}{0}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$a = 0$$

$$f(x) = 3^x$$

$$f'(x) = 3^x (1) \ln 3$$

$$f'(a) = f'(0) = 3^0 \ln 3 = (1) \ln 3 = \ln 3$$

$$\therefore \lim_{x \rightarrow 0} \frac{3^x - 1}{x} = \ln 3$$

Q#2

Find $\frac{dy}{dx}$ where

$$y = (\sin x)^{\ln x}$$

$$\ln y = \ln (\sin x)^{\ln x}$$

$$\ln y = \ln x \ln (\sin x)$$

Diff. w.r.t. x

$$\frac{y'}{y} = \frac{1}{x} \ln (\sin x) + \ln x \frac{\cos x}{\sin x}$$

* y

$$y' = y \left[\frac{1}{x} \ln (\sin x) + \ln x \cot x \right]$$

$$= (\sin x)^{\ln x} \left[\frac{1}{x} \ln (\sin x) + \ln x \cot x \right]$$

Q#3

Find an equation of tangent line P(0,1)
of

$$2y + x^2y + x \sin(x^2) = e^{xy} + 1$$

Diff. w.r.t. x

اشتق بشت + بشت اشتقا اشتق بشت + بشت اشتق

$$2y' + 2x y + x^2 y' + (1) \sin(x^2) + x \cos(x^2) 2x = e^{xy} [(1) y + x y']$$

اشتقا بشت + بشت اشتقا اشتق بشت + بشت اشتق

at P(0,1)

$$2y' + 0 + 0 + 0 + 0 = e^0 [1 + 0]$$

$$2y' = 1$$



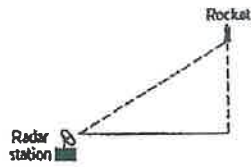
$$y' = \frac{1}{2}$$

∴ The Equation of the tangent Line:

$$y - y_0 = m(x - x_0)$$

$$y - 1 = \frac{1}{2}(x - 0)$$

Q4. [10 pts.] A rocket, rising vertically, is tracked by a radar station that is on the ground 4 km from the launchpad. How fast is the rocket rising when it is 3 km high and its distance from the radar station is increasing at a rate of 1500 km/h?

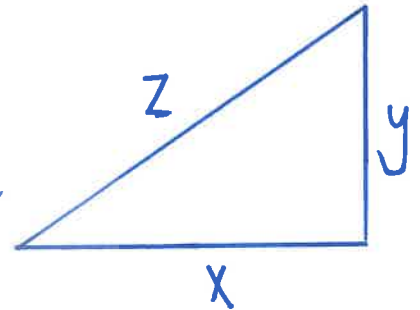


$$x = 4 \text{ km}$$

$$\frac{dy}{dt} ?? \rightarrow$$

$$y = 3 \text{ km}$$

$$\frac{dz}{dt} = 1500 \text{ km/hr}$$



$$A = \frac{1}{2}xy \quad C = x+y+z \quad Z^2 = x^2 + y^2$$

$$Z^2 = x^2 + y^2$$

$$Z^2 = (4)^2 + y^2$$

$$Z^2 = 16 + y^2$$

Diff. w.r.t. t

$$2Z \frac{dz}{dt} = 0 + 2y \frac{dy}{dt}$$

$$2(5)(1500) = 2(3) \frac{dy}{dt}$$

$$\frac{dy}{dt} = \frac{2(5)(1500)}{2(3)}$$

$$= 2500 \text{ km/hr}$$

$$Z^2 = x^2 + y^2$$

$$Z^2 = (4)^2 + (3)^2$$

$$Z^2 = 16 + 9$$

$$Z^2 = 25$$

$$Z = 5 \text{ km}$$

Q#5

Let $f(x) = e^x$

(a) Find the linearization at $a=0$.

(b) Use linear approximation to estimate $e^{0.01}$

(a) $f(x) = e^x$

$x_0 = 0$

$x_1 = x$

$\longrightarrow \Delta x = x - 0 = x$

$f(x_0) = e^0 = 1$

$f'(x) = e^x$

$f'(x_0) = e^0 = 1$

$L(x) = f(x_0) + f'(x_0) \Delta x$
 $= 1 + (1) x$

$= 1 + x$

(b) $e^{0.01} \approx 1 + 0.01$
 ≈ 1.01

Q #6

Use M.V.T to show that for
all $x > 0$

$$e^x > 1+x$$

$$f(x) = e^x \quad [0, x]$$

1) Domain of $f(x) = \mathbb{R}$ [Exponential function]
 $\therefore f(x)$ is continuous on $[0, x]$

2) $f'(x) = e^x$ [Exponential function]
 $f(x)$ is differentiable on $\mathbb{R}$
 $\therefore f(x)$ is differentiable on $(0, x)$

From ① ② $\longrightarrow \therefore$ Mean Value Theorem is satisfied
then there is a number $c \in (0, x)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$e^c = \frac{f(x) - f(0)}{x - 0}$$

$$e^c = \frac{e^x - 1}{x} \longrightarrow \textcircled{1}$$

$$0 < c < x$$
$$e^0 < e^c < e^x$$

$$1 < e^c \longrightarrow \textcircled{2}$$

$$\textcircled{1} \ln \textcircled{2}$$

$$1 < \frac{e^x - 1}{x}$$

$$x < e^x - 1$$

$$1+x < e^x$$

Q#7

$$f(x) = \frac{x^2 + 2x + 1}{x^2 - 4x + 4}$$

Domain of $f(x)$:

$$x^2 - 4x + 4 = 0$$

$$(x - 2)(x - 2) = 0$$

$$(x - 2)^2 = 0$$

$$x = 2$$

$$\Rightarrow f(x) = \frac{x^2 + 2x + 1}{(x - 2)^2}$$

$$\Rightarrow D_f = \mathbb{R} \setminus \{2\}$$

Horizontal asymptote:

$$\lim_{x \rightarrow \pm\infty} \frac{x^2 + 2x + 1}{x^2 - 4x + 4} = \lim_{x \rightarrow \pm\infty} \frac{\cancel{x^2} \left[1 + \frac{2}{x} + \frac{1}{x^2} \right]}{\cancel{x^2} \left[1 - \frac{4}{x} + \frac{4}{x^2} \right]} = \frac{1+0+0}{1-0+0}$$

$$\Rightarrow \therefore y = 1 \text{ is H.A.} = 1$$

Vertical asymptote:

$$x = 2$$

$$\lim_{x \rightarrow 2^+} \frac{x^2 + 2x + 1}{x^2 - 4x + 4} = \lim_{x \rightarrow 2^+} \frac{x^2 + 2x + 1}{(x - 2)^2} = \frac{4+4+1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 2^-} \frac{x^2 + 2x + 1}{(x - 2)^2} = \frac{4+4+1}{0^+} = +\infty$$

$$\Rightarrow \therefore x = 2 \text{ is V.A.}$$

$$f'(x) = -\frac{6(x+1)}{(x-2)^3}$$

$$= 0$$

$$-6(x+1) = 0$$

$$\div (-6)$$

$$x+1 = 0$$

$$x = -1$$

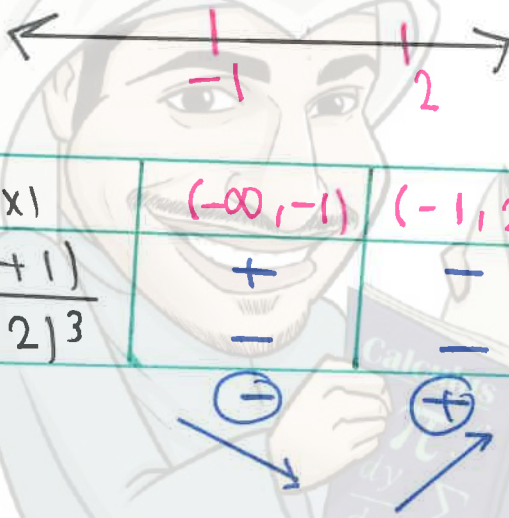
$$\in D_f$$

undefined

$$x-2=0$$

$$x=2$$

$$\notin D_f$$



$f'(x)$	$(-\infty, -1)$	$(-1, 2)$	$(2, \infty)$
$-\frac{6(x+1)}{(x-2)^3}$	+	-	-
	-	-	+
	⊖	⊕	⊖

∴ The interval of increasing $(-1, 2)$

The interval of decreasing $(-\infty, -1)$ and $(2, \infty)$

at $x = -1 \rightarrow$ L.min

$$y = \frac{1-2+1}{(-3)^2} = \frac{0}{9} = 0$$

$\Rightarrow \therefore (-1, 0)$ L.min.

$$f''(x) = \frac{6(2x+5)}{(x-2)^4}$$

$= 0$

$$6(2x+5) = 0$$

$$2x+5 = 0$$

$$2x = -5$$

$$x = \frac{-5}{2}$$

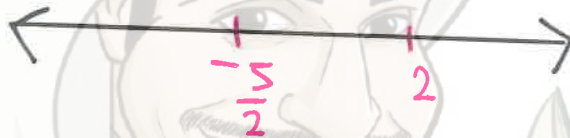
$\in D_f$

undefined

$$x-2 = 0$$

$$x = 2$$

$\notin D_f$



$f''(x)$	$(-\infty, -\frac{5}{2})$	$(-\frac{5}{2}, 2)$	$(2, \infty)$
$\frac{6(2x+5)}{(x-2)^4}$	-	+	+
	+	+	+

$\ominus$

$\oplus$

$\oplus$

$\cap$

$\cup$

$\cup$

$\therefore f(x)$ is concave up on $(-\frac{5}{2}, 2)$ and $(2, \infty)$
 $f(x)$ is concave down on $(-\infty, -\frac{5}{2})$

at $x = -\frac{5}{2} \longrightarrow$ inflection point

$$y = \frac{\frac{25}{4} + 2(-\frac{5}{2}) + 1}{(-\frac{5}{2} - 2)^2} = \frac{\frac{25}{4} - 4}{(-\frac{5}{2} - \frac{4}{2})^2} = \frac{\frac{25}{4} - \frac{16}{4}}{(-\frac{9}{2})^2} = \frac{\frac{9}{4}}{\frac{81}{4}} = \frac{1}{9}$$

$\implies \therefore (-\frac{5}{2}, \frac{1}{9})$ inf. point.

Key points

intersection point with H.A

Random point

$$1 = \frac{x^2 + 2x + 1}{x^2 - 4x + 4}$$

$$0 \mid 3 \\ 2$$

$$x^2 - 4x + 4 = x^2 + 2x + 1$$

$$\text{at } x=0 \rightarrow y = \frac{0+0+1}{0-0+4} = \frac{1}{4}$$

$$3 = 6x$$

$$x = \frac{3}{6}$$

$$x = \frac{1}{2}$$

$$\begin{aligned} \text{at } x=3 \rightarrow y &= \frac{9+6+1}{9-12+4} \\ &= \frac{16}{1} = 16 \end{aligned}$$

$\Rightarrow$

$$\left(\frac{1}{2}, 1\right)$$

$$\left(0, \frac{1}{4}\right), (3, 16)$$

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$$D_f = \mathbb{R} \setminus \{2\}$$

$$H.A \rightarrow y=1$$

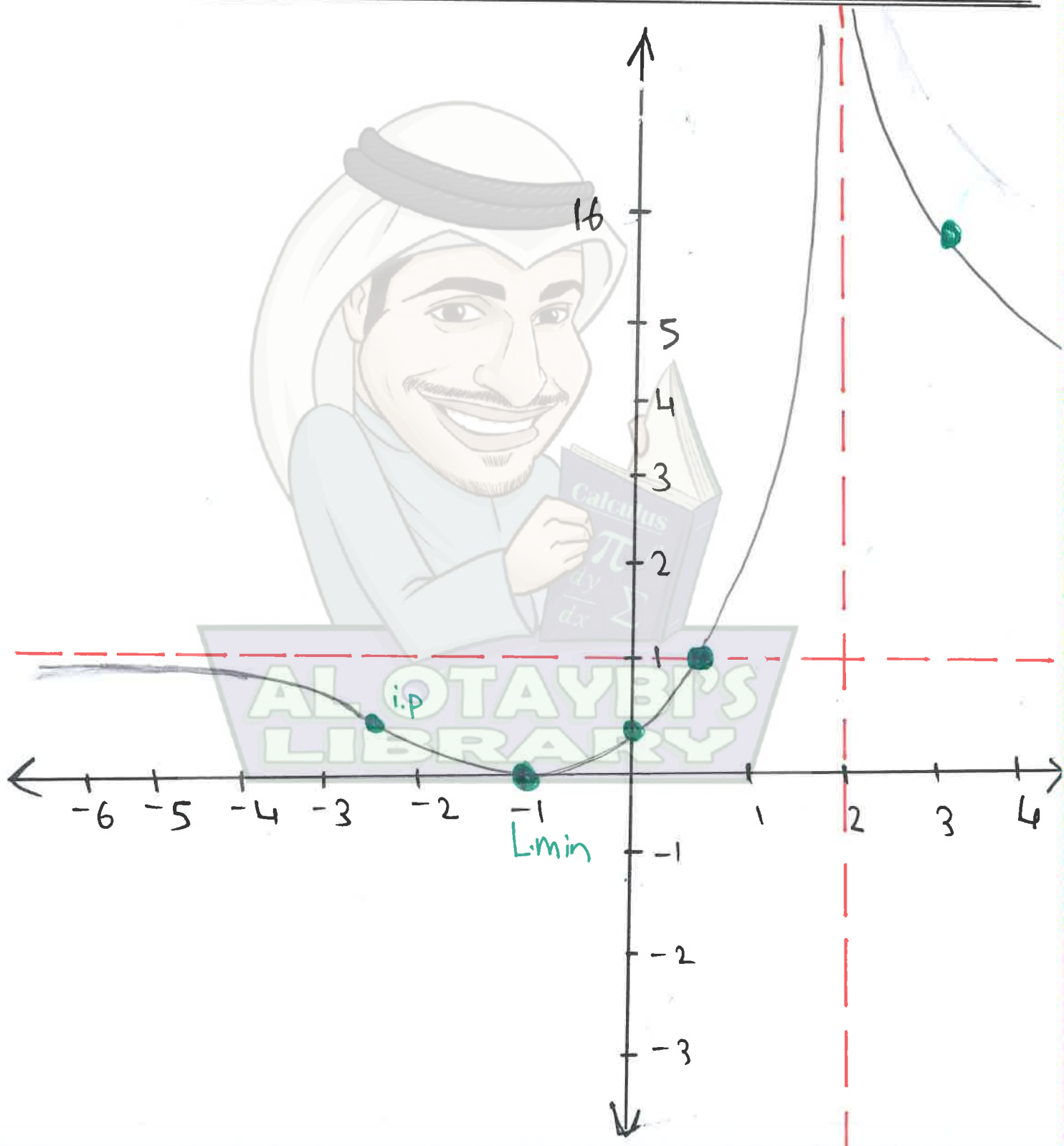
$$V.A \rightarrow x=2 \begin{cases} +\infty \\ -\infty \end{cases}$$

$$(-1, 0) \text{ L.min}$$

$$\left(-\frac{5}{2}, \frac{1}{9}\right) \text{ I.P.}$$

$$\left(-\frac{1}{2}, 1\right)$$

$$(3, 16) \quad (0, \frac{1}{4})$$





Math 101: Calculus I

Second Exam, 11 April 2019

Duration: 90 minutes

- In question 1, select the correct answer (no justification is required).
- In questions 2-9, show all your work.
- Calculators and communication devices are not allowed in the examination room.

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (20 pts)	
Q2 (10 pts)	
Q3 (10 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (10 pts)	
Q7 (10 pts)	
Q8 (10 pts)	
Q9 (10 pts)	
Total	

1. $[4 \times 5 = 20 \text{ pts.}]$ In each of the following multiple choice questions, select the correct answer.

I. The 27th derivative of the function $f(x) = \cos x - \sin x$ is

- a) $-\cos x + \sin x$.
- b) $\cos x + \sin x$.
- c) $-\cos x - \sin x$.
- d) $\cos x - \sin x$.
- e) None of the above.

II. The value of $\lim_{x \rightarrow \infty} \tanh x$ is

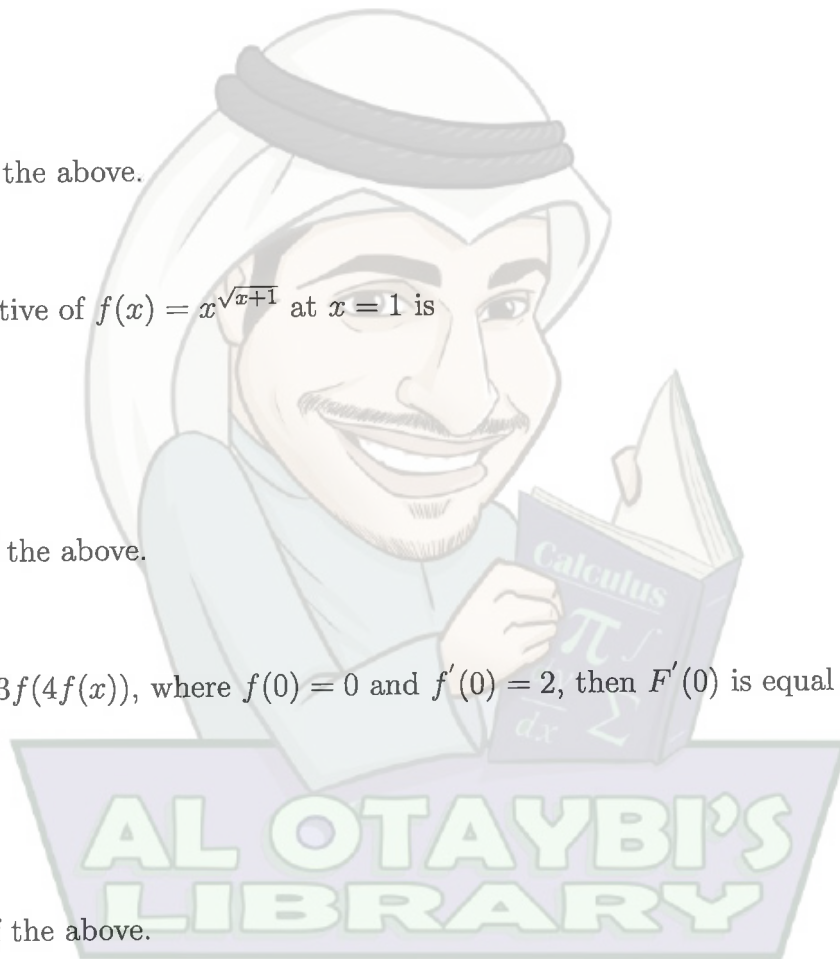
- a) does not exist.
- b) 0.
- c) $-\pi^2$.
- d) 1.
- e) None of the above.

III. The derivative of $f(x) = x^{\sqrt{x+1}}$ at $x = 1$ is

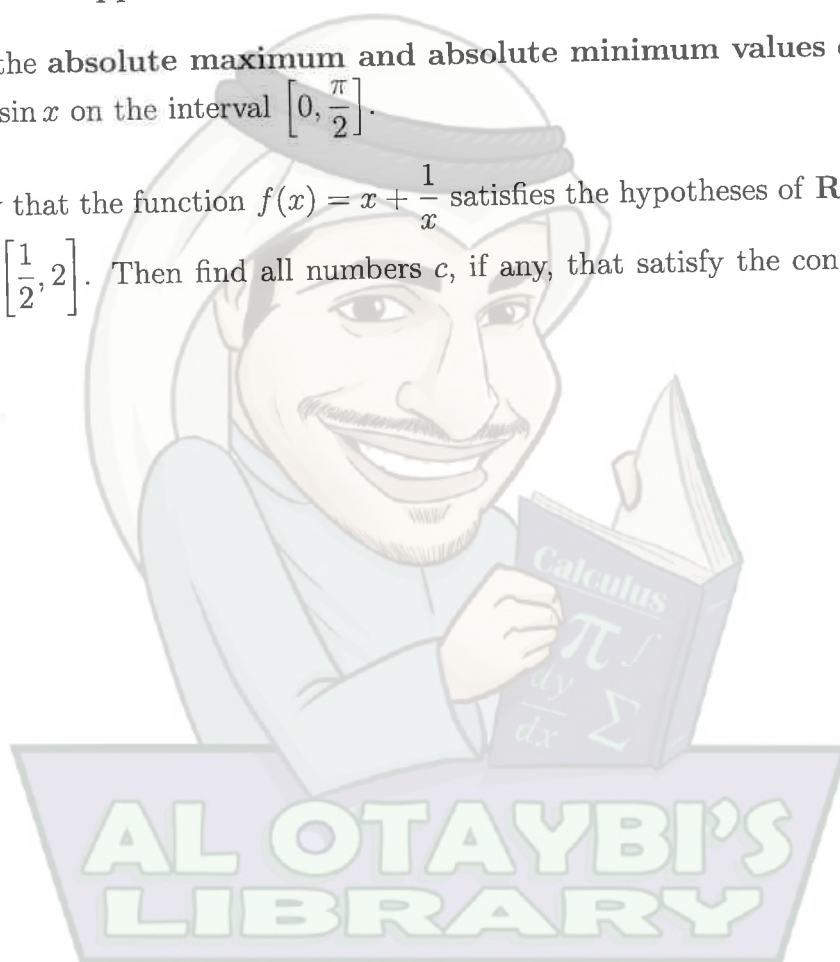
- a) 0.
- b) 1.
- c) $\sqrt{2}$.
- d) 2.
- e) None of the above.

IV. If $F(x) = 3f(4f(x))$, where $f(0) = 0$ and $f'(0) = 2$, then $F'(0)$ is equal to

- a) 48.
- b) 32.
- c) 16.
- d) 0.
- e) None of the above.



2. [10 pts.] Find **equations** of the tangent lines to the curve $f(x) = \frac{x-1}{x+1}$ that are **parallel** to the line $x - 2y = 2$.
3. [10 pts.] Evaluate the limit $\lim_{x \rightarrow 0} \frac{\sin(x^2) + 2x}{x}$, if it exists.
4. [10 pts.] Find $\frac{dy}{dx}$ by **implicit differentiation**: $e^y \sin x = x + xy$.
5. [10 pts.] Find y'' if $y = \ln(1 + \ln x)$.
6. [10 pts.] Air is being bumped into a spherical balloon so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon is increasing when the radius is 25 cm ?
7. [10 pts.] Use **linear approximation** to estimate $2e^{0.1} + (0.1)^2$.
8. [10 pts.] Find the **absolute maximum** and **absolute minimum** values of $f(x) = \cos^2 x + \sin x$ on the interval $\left[0, \frac{\pi}{2}\right]$.
9. [10 pts.] Verify that the function $f(x) = x + \frac{1}{x}$ satisfies the hypotheses of **Rolle's Theorem** on the interval $\left[\frac{1}{2}, 2\right]$. Then find all numbers c , if any, that satisfy the conclusion of Rolle's Theorem.



Q#1

I) $f(x) = \cos x - \sin x$

① $f'(x) = -\sin x - \cos x$

② $f''(x) = -\cos x + \sin x$

③ $f'''(x) = \sin x + \cos x$

④ $f^{(4)}(x) = \cos x - \sin x$

$\Rightarrow \therefore f^{(27)}(x) = \sin x + \cos x$

(b) ✓

II)

$\lim_{x \rightarrow \infty} \tanh x$

$= \lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad \frac{\infty - 0}{\infty + 0} \quad \frac{\infty}{\infty}$

$= \lim_{x \rightarrow \infty} \frac{e^x \left[1 - \frac{e^{-x}}{e^x} \right]}{e^x \left[1 + \frac{e^{-x}}{e^x} \right]}$

$= \frac{1 - 0}{1 + 0} = 1 \quad (d) \quad \checkmark$

III) $f(x) = x^{\sqrt{x+1}}$. $x=1$

$$\ln f(x) = \ln x^{\sqrt{x+1}}$$

$$\ln f(x) = \sqrt{x+1} \ln x$$

Diff. w.r.t. x

$$\frac{f'(x)}{f(x)} = \frac{1}{2\sqrt{x+1}} \ln x + \sqrt{x+1} \cdot \frac{1}{x}$$

* $f(x)$

$$f'(x) = f(x) \left[\frac{1}{2\sqrt{x+1}} \ln x + \sqrt{x+1} \cdot \frac{1}{x} \right]$$

$$= x^{\sqrt{x+1}} \left[\frac{1}{2\sqrt{x+1}} \ln x + \sqrt{x+1} \cdot \frac{1}{x} \right]$$

at $x=1$

$$f'(1) = 1^{\pi} \left[0 + \sqrt{2} \cdot \frac{1}{1} \right]$$

$$= \sqrt{2}$$

(C) ✓

IV) If $F(x) = 3f(4f(x))$. $f(0) = 0$
 $f'(0) = 2$
 $F'(0) ??$

$$F'(x) = 3f'(4f(x)) \cdot 4f'(x)$$

at $x=0$

$$F'(0) = 3f'(4f(0)) \cdot 4f'(0)$$

$$= 3f'(4(0)) \cdot 4(2)$$

$$= 3f'(0) \cdot 8$$

$$= 3(2) \cdot 8$$

$$= 48 \quad (a)$$

Q#2

Find the equation of tangent lines to

$$f(x) = \frac{x-1}{x+1} \quad || \quad x-2y=2$$

$$f(x) = \frac{x-1}{x+1}$$

$$\begin{aligned} f'(x) &= \frac{(x+1)(1) - (x-1)(1)}{(x+1)^2} \\ &= \frac{x+1 - x+1}{(x+1)^2} \\ &= \frac{2}{(x+1)^2} \end{aligned}$$

$$x-2y=2$$

Diff. w.r.t. x

$$1 - 2y' = 0$$

$$2y' = 1$$

$$y' = \frac{1}{2}$$

∴ The tangent line of f is parallel to the line y

$$\therefore f'(x) = y'$$

$$\frac{2}{(x+1)^2} = \frac{1}{2}$$

$$4 = (x+1)^2$$

$$x+1=2$$

$$\boxed{x=1}$$

$$x+1=-2$$

$$\boxed{x=-3}$$

$$\text{at } x=1 \longrightarrow y = \frac{0}{2} = 0$$

$$\text{at } x=-3 \longrightarrow y = \frac{-4}{-2} = 2$$

$$(1, 0) \quad (-3, 2)$$

∴ The equation of tangent lines

$$y - y_0 = m(x - x_0)$$

at $(1, 0)$

$$y - 0 = \frac{1}{2}(x - 1)$$

at $(-3, 2)$

$$y - 2 = \frac{1}{2}(x + 3)$$

Q#3

$$\lim_{x \rightarrow 0} \frac{\sin(x^2) + 2x}{x}$$

$\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{x^2 \frac{\sin(x^2)}{x^2} + 2x}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{x} \left[\cancel{x} \frac{\sin(x^2)}{x^2} + 2 \right]}{\cancel{x}}$$

$$= \frac{0(1) + 2}{1} = 0 + 2 = 2$$

Sol #2

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$a = 0$$

$$f(x) = \sin(x^2) + 2x$$

$$f'(x) = \cos(x^2)(2x) + 2$$

$$f'(a) = f'(0) = 0 + 2 = 2$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin(x^2) + 2x}{x} = 2$$

Q#4

Find $\frac{dy}{dx}$

$$e^y \sin x = x + x y$$

Diff. w.r. t. x

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$$e^y y' \sin x + e^y \cos x = 1 + (1) y + x y'$$

$$e^y y' \sin x - x y' = 1 + y - e^y \cos x$$

$$y' [e^y \sin x - x] = 1 + y - e^y \cos x$$

$$\div [e^y \sin x - x]$$

$$y' = \frac{1 + y - e^y \cos x}{e^y \sin x - x}$$

Q #5

Find y''

$$y = \ln(1 + \ln x)$$

$$y' = \frac{\frac{1}{x}}{1 + \ln x}$$

$$y' = \frac{1}{x(1 + \ln x)}$$

$$y'' = \frac{x(1 + \ln x) \cdot 0 - 1 \cdot [x(1 + \ln x)]'}{[x(1 + \ln x)]^2}$$

$$= \frac{0 - [(1)(1 + \ln x) + x \cdot \frac{1}{x}]}{[x(1 + \ln x)]^2}$$

Q#6

$$\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$$

$$\frac{dr}{dt} ?? \rightarrow r = 25 \text{ cm}$$



$$V = \frac{4}{3}\pi r^3 \quad S = 4\pi r^2$$

Diff. w.r.t. t

$$\frac{dV}{dt} = \frac{4}{3} \pi 3r^2 \frac{dr}{dt}$$

$$100 = 4\pi (25)^2 \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = \frac{100}{4\pi (25)^2}$$

$$= \frac{25}{\pi (25)^2}$$

$$= \frac{1}{\pi (25)} \text{ cm/s}$$

Q# 7

$$\underline{2e^{0.1} + (0.1)^2}$$

$$f(x) = 2e^x + x^2$$

$$x_0 = 0$$

$$x_1 = 0.1$$

$$\longrightarrow \Delta x = 0.1 - 0 = 0.1$$

$$1) f(x_0) = 2e^0 + 0 = 2(1) = \boxed{2}$$

$$2) f'(x) = 2e^x + 2x$$

$$3) f'(x_0) = 2e^0 + 0 = 2(1) = \boxed{2}$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 2 + 2(0.1)$$

$$\approx 2 + 0.2$$

$$\approx 2.2$$

Q #8

$$f(x) = \cos^2 x + \sin x$$

$$\left[0, \frac{\pi}{2}\right]$$

$$f'(x) = 2\cos x (-\sin x) + \cos x$$

$$= -2\cos x \sin x + \cos x$$

$$= 0$$

undefined
None

$$-2\cos x \sin x + \cos x = 0$$

$$\cos x [-2\sin x + 1] = 0$$

$$\cos x = 0$$

$$-2\sin x + 1 = 0$$

$$\div 2$$

$$2\sin x = 1$$

$$\sin x = \frac{1}{2}$$

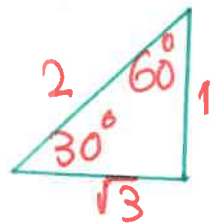
$$x = \frac{\pi}{6} \in \left[0, \frac{\pi}{2}\right]$$

$$x = \frac{\pi}{2}$$

$$\in \left[0, \frac{\pi}{2}\right]$$

$$x = \frac{3\pi}{2}$$

$$\notin \left[0, \frac{\pi}{2}\right]$$



$$f\left(\frac{\pi}{6}\right) = \cos^2 \frac{\pi}{6} + \sin \frac{\pi}{6} = \left(\frac{\sqrt{3}}{2}\right)^2 + \frac{1}{2}$$

$$= \frac{3}{4} + \frac{1}{2} = \frac{3}{4} + \frac{2}{4} = \frac{5}{4}$$

$$f(0) = (1)^2 + 0 = 1$$

$$f\left(\frac{\pi}{2}\right) = 0 + \sin \frac{\pi}{2} = 1$$

∴ The absolute maximum value = $\frac{5}{4}$
The absolute minimum value = 1

Q# 9 $f(x) = x + \frac{1}{x}$ $[\frac{1}{2}, 2]$ Rolle's Thm?

1) Domain of $f(x) = \mathbb{R} \setminus \{0\}$ [$\frac{1}{x}$: reciprocal function]
 $\therefore 0 \notin [\frac{1}{2}, 2]$ (~~rational function~~)

$\therefore f(x)$ is continuous on $[\frac{1}{2}, 2]$

2) $f'(x) = 1 - \frac{1}{x^2}$ [Rational function]

$f(x)$ is differentiable on $\mathbb{R} \setminus \{0\}$

$\therefore 0 \notin (\frac{1}{2}, 2)$

$\therefore f(x)$ is differentiable on $(\frac{1}{2}, 2)$

3) $f(\frac{1}{2}) = \frac{1}{2} + \frac{1}{\frac{1}{2}} = \frac{1}{2} + 2$

$f(2) = 2 + \frac{1}{2} \longrightarrow \therefore f(\frac{1}{2}) = f(2)$

From ① ② ③ $\longrightarrow \therefore$ Rolle's Theorem is satisfied
then there is a number $c \in (\frac{1}{2}, 2)$ such that

$f'(c) = 0$

$1 - \frac{1}{c^2} = 0$

$1 = \frac{1}{c^2}$

$c^2 = 1$

$c = -1 \notin (\frac{1}{2}, 2)$

$c = 1 \in (\frac{1}{2}, 2)$

Kuwait University
Department of Mathematics

6 May 2017

Calculus I (Math 101)

Duration 90 minutes

Second Exam

Answer all questions. Calculators and mobile phones are not allowed.

1. [10 pts.] Let $y = (x^2 + 1)^{\cos x}$. Use logarithmic differentiation to find $\frac{dy}{dx}$ at $x = 0$.
2. [10 pts.] Two people start running from the same point P . One runs east at 6 km/h and the other runs north at 8 km/h. How fast is the distance between the two people changing after 0.5 hour?
3. [10 pts.] Let $f(x) = \tan^{-1}(x) - \frac{\pi}{4}$. Use linear approximation to estimate $f(1.05)$.
4. [10 pts.] Find the absolute maximum and absolute minimum values of $f(x) = x^2 e^{-2x}$ on the interval $[-1, 1]$.
5. [10 × 4 = 40 pts.] Let $f(x) = \ln(x^2 + 1)$.
 - (a) i) Study the symmetry of the graph of f .
 - ii) Find the horizontal asymptotes of the graph of f , if any.
 - (b) Given that $f'(x) = \frac{2x}{x^2 + 1}$.
 - i) Find the intervals on which f is increasing and the intervals on which f is decreasing, if any.
 - ii) Find the local maximum and minimum values of f , if any.
 - (c) Given that $f''(x) = \frac{2(1 - x^2)}{(x^2 + 1)^2}$.
 - i) Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward, if any.
 - ii) Find the points of inflection, if any.
 - (d) Sketch the graph of f .
6. [5 × 4 = 20 pts.] In each of the following multiple choice questions, select the correct answer.

- I. $\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x^2} =$
 - a) 1.
 - b) 0.
 - c) -1.
 - d) ∞ .
 - e) None of the above.
- II. If $f(x) = 5x$ and $g(x) = 2 \cos x$, then $(f \circ g)'(\pi) =$
 - a) -5.
 - b) 0.
 - c) 10.
 - d) -10.
 - e) None of the above.
- III. The function f defined by $f(x) = \frac{1}{x}$ has a critical point at:
 - a) $x = 0$.
 - b) $x = 1$.
 - c) $x = -1$.
 - d) $x = 2$.
 - e) None of the above.
- IV. If $\sin(x) + y^3 = x^2 + 1$, then $\left. \frac{dy}{dx} \right|_{(0,1)} =$
 - a) $-1/3$.
 - b) $1/3$.
 - c) 0.
 - d) 1.
 - e) None of the above.

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1) $y = (x^2 + 1)^{\cos x}$. Find $\frac{dy}{dx}$ at $x=0$.

$$\ln y = \ln (x^2 + 1)^{\cos x}$$

$$\ln y = \cos x \ln (x^2 + 1)$$

Diff. w.r. to x

الشتق بـ x + مشتق الاستيعا

$$\frac{y'}{y} = -\sin x \ln (x^2 + 1) + \cos x \frac{2x}{x^2 + 1}$$

* y

$$y' = y \left[-\sin x \ln (x^2 + 1) + \cos x \frac{2x}{x^2 + 1} \right]$$
$$= (x^2 + 1)^{\cos x} \left[-\sin x \ln (x^2 + 1) + \cos x \frac{2x}{x^2 + 1} \right]$$

at $x=0$

$$y' = (0+1)^{(1)} [0 + (1)0]$$

$$y' = (1) [0]$$

$$y' = 0$$

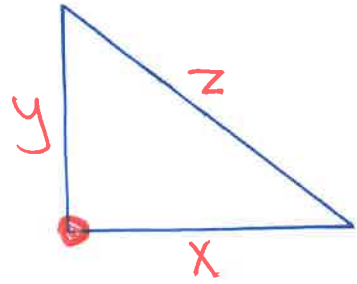


②

$$\frac{dx}{dt} = 6 \text{ km/hr}$$

$$\frac{dy}{dt} = 8 \text{ km/hr}$$

$$\frac{dz}{dt} ?? \rightarrow t = 0.5 \text{ hr}$$



$$z^2 = x^2 + y^2$$

Diff. w.r. t. time.

$$2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

$$x = \frac{dx}{dt} * t$$

$$= 6 (0.5) = 3 \text{ km}$$

$$2 (5) \frac{dz}{dt} = 2(3)(6) + 2(4)(8)$$

$$y = \frac{dy}{dt} * t$$

$$10 \frac{dz}{dt} = 36 + 64$$

$$= 8 (0.5) = 4 \text{ km}$$

$$10 \frac{dz}{dt} = 100$$

$$z^2 = x^2 + y^2$$

$$z^2 = (3)^2 + (4)^2$$

$$z^2 = 9 + 16$$

$$z^2 = 25$$

$$z = 5 \text{ km}$$

$$\frac{dz}{dt} = \frac{100}{10} = 10 \text{ km/hr}$$

$$\textcircled{3} \quad f(x) = \tan^{-1}x - \frac{\pi}{4}$$

$$x_0 = 1$$

$$x_1 = 1.05 \longrightarrow \Delta x = 1.05 - 1 = 0.05$$

$$f(x_0) = \tan^{-1}1 - \frac{\pi}{4} = \frac{\pi}{4} - \frac{\pi}{4} = 0$$

$$f'(x) = \frac{1}{x^2 + 1}$$

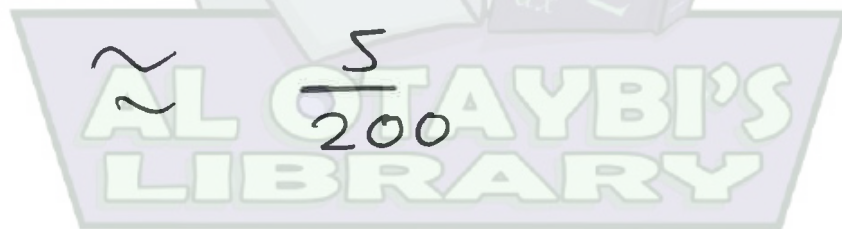
$$f'(x_0) = \frac{1}{(1)^2 + 1} = \frac{1}{1+1} = \frac{1}{2}$$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 0 + \frac{1}{2} (0.05)$$

$$\approx \frac{1}{2} \frac{5}{100}$$

$$\approx \frac{5}{200}$$



④ $f(x) = x^2 e^{-2x}$ $[-1, 1]$

Domain of $f(x) = \mathbb{R}$ [polynomial of $d=2$
exponential]

$$\begin{aligned} f'(x) &= 2x e^{-2x} + x^2 e^{-2x} (-2) \\ &= 2x e^{-2x} - 2x^2 e^{-2x} \\ &= x e^{-2x} [2 - x] \end{aligned}$$

$x e^{-2x} [2 - x] = 0$ DNE
None

$x = 0$ $e^{-2x} = 0$ $2 - x = 0$
 $\in$ None $x = 2$
 $[-1, 1]$ $\notin (-1, 1)$

$f(-1) = 1e^2 = e^2$

$f(1) = 1e^{-2} = \frac{1}{e^2}$

$f(0) = 0$

$\therefore$ absolute maximum is e^2

absolute minimum is 0

Q#5 Domain of $f(x)$: $x^2 + 1 > 0$
 $x^2 > -1$ (always)

$$f(x) = \ln(x^2 + 1)$$

$$\Rightarrow D_f = \mathbb{R}$$

Horizontal asymptote

$$\lim_{x \rightarrow \pm\infty} \ln(x^2 + 1) = \ln\infty = \infty$$

$\Rightarrow \therefore$ No HA

Vertical asymptote

No V.A

$$f'(x) = \frac{2x}{x^2 + 1}$$

$$= 0$$

$$2x = 0$$

$\div 2$

$$x = 0 \in D_f$$

undefined

$$x^2 + 1 = 0$$

$$x^2 = -1$$

None

$$-\infty \longleftarrow \begin{array}{c} | \\ 0 \end{array} \longrightarrow \infty$$

$f'(x)$	$(-\infty, 0)$	$(0, \infty)$
$\frac{2x}{x^2 + 1}$	-	+
	$\ominus$	$\ominus$

$\therefore$ The interval of decreasing = $(-\infty, 0)$ and $(0, \infty)$
 No interval of increasing / No Local extrema.

$$f''(x) = \frac{2(1-x^2)}{(x^2+1)^2}$$

$$= 0$$

$$2(1-x^2) = 0$$

$$\div 2 \quad 1-x^2 = 0$$

$$(1-x)(1+x) = 0$$

$$x = 1 \quad x = -1$$

EDP

EDP

undefined

$$x^2 + 1 = 0$$

$$x^2 = -1$$

None



$f''(x)$	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
$\frac{2(1-x)(1+x)}{(x^2+1)^2}$	-	+	-
	+	+	+



∴ the interval of concave upward = $(-1, 1)$

the interval of concave downward = $(-\infty, -1)$ and $(1, \infty)$

∴ at $x = -1 \rightarrow y = \ln(1+1) = \ln 2$

at $x = 1 \rightarrow y = \ln(1+1) = \ln 2$

∴ $(-1, \ln 2)$ $(1, \ln 2)$ are inflection points

key points

Random point

at $x = 2 \rightarrow y = \ln(4+1) = \ln 5$

$\Rightarrow P(2, \ln 5)$

$D_f = \mathbb{R}$

HA $\rightarrow$ None

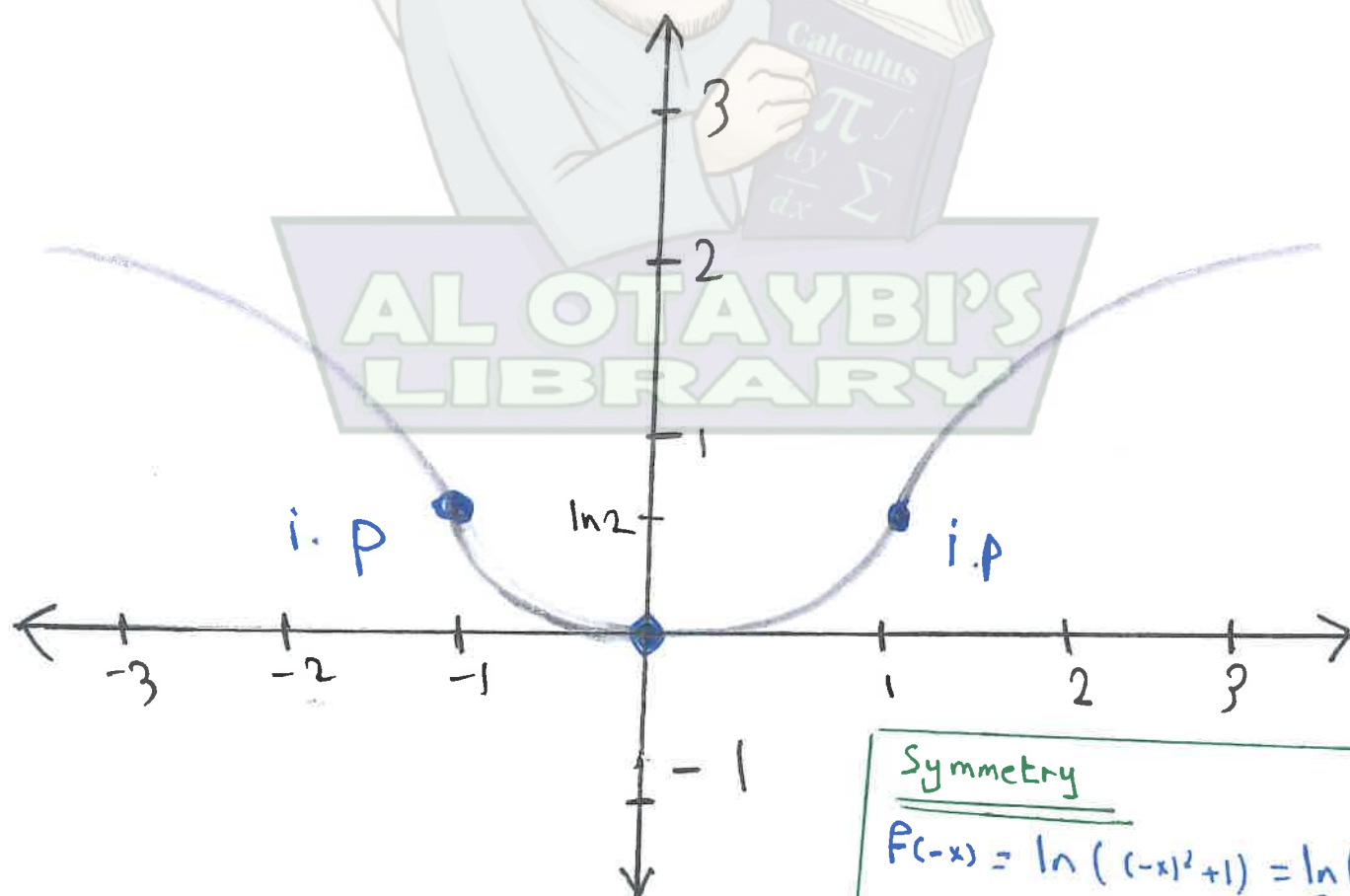
V.A $\rightarrow$ None

(0,0) L.min

$(-1, \ln 2)$ $(1, \ln 2)$ inflection points.

$\cup$ $(-1, 1)$ $(-\infty, -1)$ $(1, \infty)$

$(2, \ln 5)$



Symmetry

$$f(-x) = \ln((-x)^2 + 1) = \ln(x^2 + 1) = f(x)$$

$\therefore$ The symmetry about y-axis.

multiple choice

I) $\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x^2} =$

- a) 1 b) 0 c) -1 d) ∞ e) none of the above

Sol

$$\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{-\sin^2 x}{x^2}$$

$$\lim_{x \rightarrow 0} - \frac{\sin x \sin x}{x^2}$$

$$\lim_{x \rightarrow 0} - \frac{\cancel{x} \frac{\sin x}{x} \cancel{x} \frac{\sin x}{x}}{x^2} = - (1)(1) = -1$$

(c) ✓

II) Video

III) The function f defined by $f(x) = \frac{1}{x}$ has a critical point at :

- a) $x=0$ b) $x=1$ c) $x=-1$ d) $x=2$ e) none of the above

Sol

$$f(x) = \frac{1}{x}$$

Domain of $f(x) = \mathbb{R} \setminus \{0\}$

$$f'(x) = \frac{-1}{x^2}$$

$$= 0$$

$$-1 = 0$$

None

undefined

$$x^2 = 0$$

$$x = 0$$

⊄ Domain

∴ No critical point

(e) ✓

IV) If $\sin x + y^3 = x^2 + 1$ then $\frac{dy}{dx} \Big|_{(0,1)} =$

- a) $-\frac{1}{3}$ b) $\frac{1}{3}$ c) 0 d) 1 e) none of the above

sol

$$\sin x + y^3 = x^2 + 1$$

Diff. w.r.t. x

$$\cos x + 3y^2 y' = 2x$$

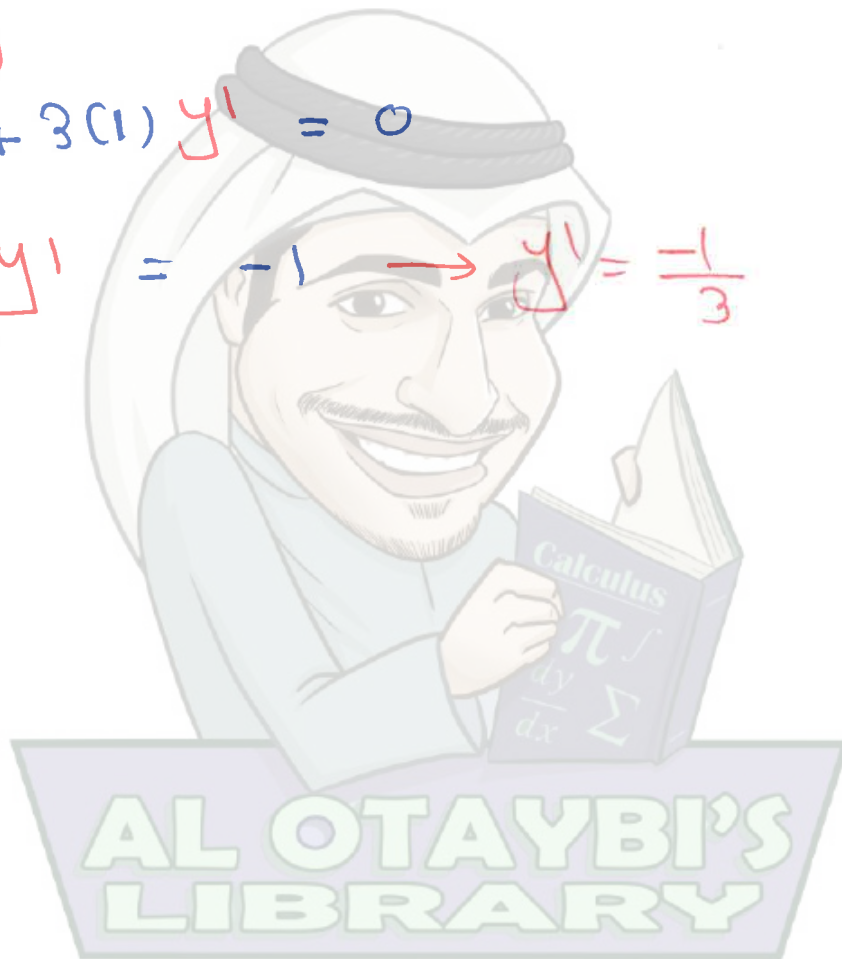
at $(0,1)$

$$\cos(0) + 3(1)y' = 0$$

↙

$$3y' = -1 \rightarrow y' = -\frac{1}{3}$$

Ⓐ ✓



Kuwait University
Department of Mathematics

Calculus I (Math 101)

22 July 2017

Second Exam

Duration 90 minutes

Answer all questions. Calculators and mobile phones are not allowed.

1. [$5 \times 4 = 20$ pts.] In each of the following multiple choice questions, select the correct answer.

I. If $f(x) = \sin^{-1}(2x)$, then $f'(x) =$

(A) $\frac{1}{\sqrt{1-4x^2}}$.

(B) $\frac{-1}{\sqrt{1-4x^2}}$.

(C) $\frac{2}{\sqrt{1-4x^2}}$.

(D) $\frac{-2}{\sqrt{1-4x^2}}$.

(E) None of the above.

II. The slope of the tangent line to the graph of $x^2 - 3y^3 = -2$ at $(1, 1)$ is

(A) $1/3$.

(B) $-9/2$.

(C) $2/9$.

(D) $3/2$.

(E) None of the above.

III. $\lim_{x \rightarrow 0} x \cot(\pi x) =$

(A) π .

(B) $-\pi$.

(C) $\frac{1}{\pi}$.

(D) $-\frac{1}{\pi}$.

(E) None of the above.

IV. The derivative of $\cosh^2 x - \sinh^2 x$ is

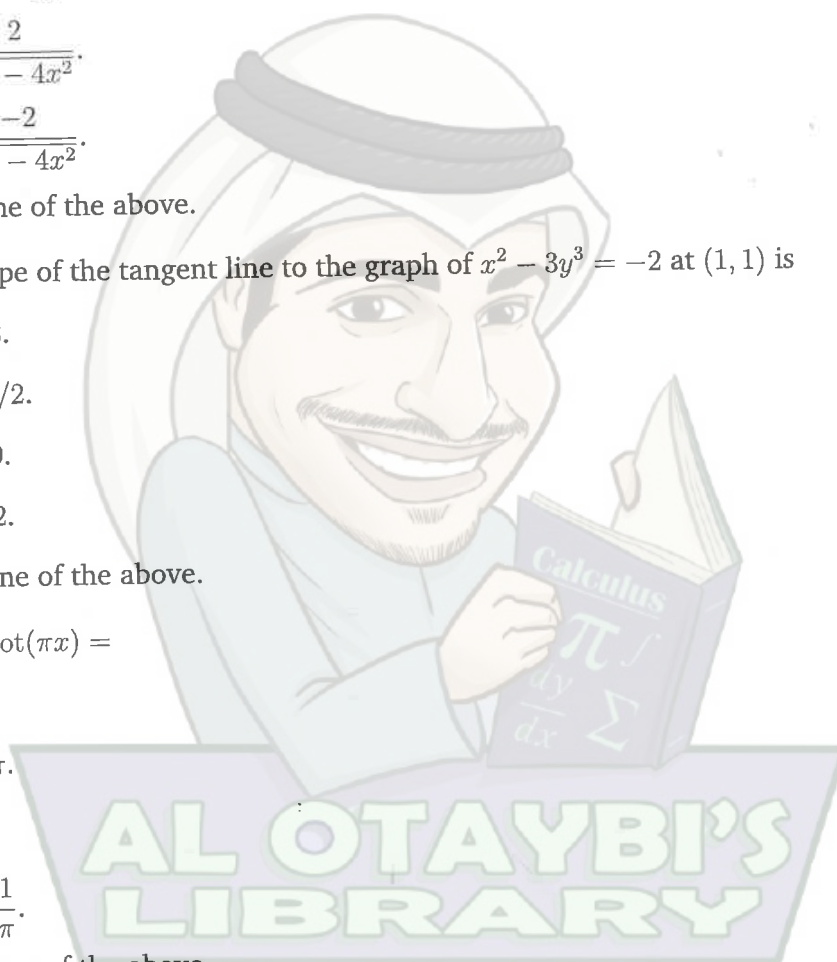
(A) 0.

(B) $4 \cosh x \sinh x$.

(C) $-2 \cosh x \sinh x$.

(D) $2 \cosh x \sinh x$.

(E) None of the above.



2. [10 pts.] Use the definition of the derivative to show that the function $f(x) = \frac{1}{x-1}$ is differentiable at $x = 2$.
3. [10 pts.] Let $f(x) = (3 + 4x)^{\cos x}$. Find $f'(x)$.
4. [10 pts.] Verify that $f(x) = \sqrt{4 - x^2}$ satisfies the hypotheses of the Mean Value Theorem on the interval $[-2, 0]$. Then, find all numbers c that satisfy the conclusion of the Theorem.
5. [10 pts.] Find the linearization of the function $f(x) = x \ln x$ at $a = 1$ and use it to approximate $f(1.1)$.
6. [$4 \times 10 = 40$ pts.] Let $f(x) = \frac{x^2 - x}{(x + 1)^2}$.

(a) Find the vertical and horizontal asymptotes of the graph of f , if any.

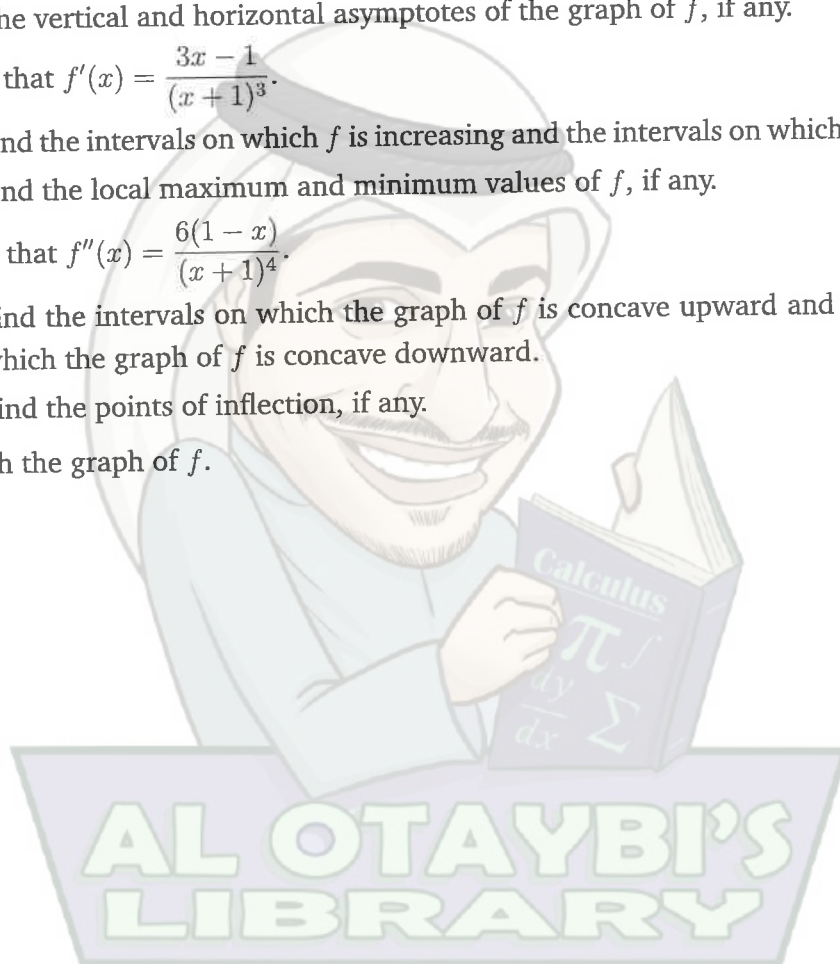
(b) Given that $f'(x) = \frac{3x - 1}{(x + 1)^3}$.

- Find the intervals on which f is increasing and the intervals on which f is decreasing.
- Find the local maximum and minimum values of f , if any.

(c) Given that $f''(x) = \frac{6(1 - x)}{(x + 1)^4}$.

- Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward.
- Find the points of inflection, if any.

(d) Sketch the graph of f .



Q#1

I) $f(x) = \sin^{-1}(2x)$, then $f'(x)$.

$$f'(x) = \frac{2}{\sqrt{1-(2x)^2}} = \frac{2}{\sqrt{1-4x^2}}$$

(c) ✓

II) $x^2 - 3y^3 = -2$ at (1,1) slope of tan.lie.

Diff. w.r.t.x

$$2x - 9y^2 y' = 0$$

at (1,1)

$$2(1) - 9(1) y' = 0$$

$$2 = 9 y'$$

$\div 9$

$$y' = \frac{2}{9} \quad (c) \checkmark$$

III) $\lim_{x \rightarrow 0} x \cot(\pi x)$

$$= \lim_{x \rightarrow 0} x \frac{1}{\tan(\pi x)}$$

$$= \lim_{x \rightarrow 0} \frac{x}{\tan(\pi x)}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{x}}{\cancel{\pi x} \tan(\pi x)}$$

$$= \frac{1}{\pi (1)} = \frac{1}{\pi} \quad (C) \checkmark$$

IV) The derivative of $(\cosh^2 x - \sinh^2 x)$

$$= 2\cosh x \sinh x - 2\sinh x \cosh x = 0$$

SOL #2

$$(\cosh^2 x - \sinh^2 x)'$$

$$= (1)'$$

$$= 0 \quad (a) \checkmark$$

Q) Use definition of derivative to show that
 #2) $f(x) = \frac{1}{x-1}$ is differentiable at $x=2$

اقرأه عن
 المساحة
 2.8

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{x \rightarrow 2} \frac{\frac{1}{x-1} - f(2)}{x-2} \rightarrow f(2) = \frac{1}{2-1} = \frac{1}{1} = 1$$

$$= \lim_{x \rightarrow 2} \frac{\frac{1}{x-1} - 1}{x-2}$$

$$= \lim_{x \rightarrow 2} \frac{\frac{1}{x-1} - \frac{x-1}{x-1}}{x-2}$$

$$= \lim_{x \rightarrow 2} \frac{1 - (x-1)}{(x-1)(x-2)}$$

$$= \lim_{x \rightarrow 2} \frac{1 - x + 1}{(x-1)(x-2)}$$

$$= \lim_{x \rightarrow 2} \frac{2 - x}{(x-1)(x-2)}$$

$$= \lim_{x \rightarrow 2} \frac{-(x-2)}{(x-1)(x-2)} = \frac{-1}{2-1}$$

$$\therefore f \text{ is differentiable at } -1 = \frac{-1}{1} = -1$$

③ $F(x) = (3+4x)^{\cos x}$. Find $F'(x)$

$$F(x) = (3+4x)^{\cos x}$$

$$\ln F(x) = \ln (3+4x)^{\cos x}$$

$$\ln F(x) = \cos x \ln (3+4x)$$

Diff. w.r.t. x

$\frac{F'(x)}{F(x)} =$
الشيء الثاني + الشيء الأول

$$= -\sin x \ln (3+4x) + \cos x \frac{4}{3+4x}$$

* $F(x)$

$$F'(x) = F(x) \left[-\sin x \ln (3+4x) + \cos x \frac{4}{3+4x} \right]$$

$$= (3+4x)^{\cos x} \left[-\sin x \ln (3+4x) + \cos x \frac{4}{3+4x} \right]$$

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Q#4

$f(x) = \sqrt{4-x^2}$

$[-2, 0]$

M.V.T?

1) Domain of $f(x)$:

$$\begin{aligned} 4 - x^2 &\geq 0 \\ 4 &\geq x^2 \\ [-2, 2] \end{aligned}$$

$$\Rightarrow \therefore D_f = [-2, 2]$$

$\Rightarrow \therefore f(x)$ is continuous on $[-2, 0]$

2) $f'(x) = \frac{-2x}{2\sqrt{4-x^2}}$

$f(x)$ is differentiable on $(-2, 2)$

$\therefore f(x)$ is differentiable on $(-2, 0)$

From ①② $\rightarrow \therefore$ Mean value Theorem is satisfied
then there is a number $c \in (-2, 0)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\frac{-c}{\sqrt{4-c^2}} = \frac{f(0) - f(-2)}{0 - (-2)}$$

$$\frac{-c}{\sqrt{4-c^2}} = \frac{\sqrt{4-0} - \sqrt{4-4}}{2} \rightarrow 0$$

$$\frac{-c}{\sqrt{4-c^2}} = \frac{2}{2}$$

$$-c = \sqrt{4-c^2}$$

$$(-c)^2 = (\sqrt{4-c^2})^2$$

$$\begin{aligned} c^2 &= 4 - c^2 \\ 2c^2 &= 4 \\ \div 2 \\ c^2 &= 2 \end{aligned}$$

$$\boxed{c = -\sqrt{2}} \\ \in (-2, 0)$$

$$c = \sqrt{2} \\ \notin (-2, 0)$$

Q#5

$f(x) = x \ln x$ at $a=1$ (a) Linearization

(b) $f(1.1)$

$$f(x) = x \ln x$$

$$x_0 = 1$$

$$x_1 = x$$

$$\longrightarrow \Delta x = x - 1$$

$$f(x_0) = (1) \ln 1 = \boxed{0}$$

$$f'(x) = (1) \ln x + x \cdot \frac{1}{x}$$

$$= \ln x + 1$$

$$f'(x_0) = \ln 1 + 1 = 0 + 1 = \boxed{1}$$

$$\begin{aligned} L(x) &= f(x_0) + f'(x_0) \Delta x \\ &= 0 + 1 \cdot (x - 1) \\ &= x - 1 \end{aligned}$$

$$\begin{aligned} f(1.1) &\approx 1.1 - 1 \\ &\approx 0.1 \end{aligned}$$

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$$f(x) = \frac{x^2 - x}{(x+1)^2}$$

Q#6

① Domain of $f(x)$: $x+1=0$
 $x=-1$

$$\Rightarrow \therefore \text{Df} = \mathbb{R} \setminus \{-1\}$$

② H.A $\lim_{x \rightarrow \pm\infty} \frac{x^2 - x}{(x+1)^2}$

$$\lim_{x \rightarrow \pm\infty} \frac{x^2(1 - \frac{1}{x})}{[x(1 + \frac{1}{x})]^2}$$

$$\lim_{x \rightarrow \pm\infty} \frac{x^2(1 - \frac{1}{x})}{x^2(1 + \frac{1}{x})^2} = \frac{1-0}{(1+0)^2} = 1$$

$\therefore y=1$ is H.A

③ V.A $x=-1$

$$\lim_{x \rightarrow -1^+} \frac{x^2 - x}{(x+1)^2} = \frac{1+1}{0^+} = +\infty$$

$$\lim_{x \rightarrow -1^-} \frac{x^2 - x}{(x+1)^2} = \frac{1+1}{0^-} = +\infty$$

$$f'(x) = \frac{3x-1}{(x+1)^3}$$

= 0

$$3x - 1 = 0$$

$$3x = 1$$

$$x = \frac{1}{3}$$

$\in D_f$

undefined (D.N.E)

$$x+1=0$$

$$x = -1$$

$\notin D_f$



$f'(x)$	$(-\infty, -1)$	$(-1, \frac{1}{3})$	$(\frac{1}{3}, \infty)$
$\frac{3x-1}{(x+1)^3}$	-	-	+
	-	+	+

(+)

(-)

(+)

$\therefore$ interval of increasing are $(-\infty, -1)$ and $(\frac{1}{3}, \infty)$
interval of decreasing $(-1, \frac{1}{3})$

at $x = \frac{1}{3} \longrightarrow$ L. min

$$y = \frac{\frac{1}{3} - \frac{1}{3}}{(\frac{1}{3} + 1)^2} = \frac{\frac{1}{3} - \frac{2}{3}}{(\frac{1}{3} + \frac{2}{3})^2} = \frac{\frac{-2}{3}}{(\frac{4}{3})^2} = \frac{\frac{-2}{3}}{\frac{16}{9}} = \frac{-2}{16} = -\frac{1}{8}$$

$\Rightarrow \boxed{(\frac{1}{3}, -\frac{1}{8}) \text{ L. min}}$

$$f''(x) = \frac{6(1-x)}{(x+1)^4}$$

$= 0$

$$6(1-x) = 0$$

$\div 6$

$$1-x = 0$$

$$x = 1$$

$\in \text{Domain}$

DNE

$$x+1=0$$

$$x = -1$$

$\notin \text{Domain}$

$f''(x)$	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
$\frac{6(1-x)}{(x+1)^4}$	+	+	-
	+	+	+
	⊕	⊕	⊖
	∪	∪	∩

$\therefore f(x)$ is concave up on $(-\infty, -1)$ $(-1, 1)$

$f(x)$ is concave down on $(1, \infty)$

as at

$x=1 \rightarrow$ inflection point

$$y = \frac{1-1}{(2)^2} = \frac{0}{4} = 0$$

$(1, 0)$ I.P

Key points

$$1 = \frac{x^2 - x}{(x+1)^2}$$

$$(x+1)^2 = x^2 - x$$

$$x^2 + 2x + 1 = x^2 - x$$

$$x^2 + 2x + 1 - x^2 + x = 0$$

$$3x + 1 = 0$$

$$3x = -1$$

$$x = -\frac{1}{3}$$

$$\left(-\frac{1}{3}, 1\right)$$

$$-2 \parallel 0$$
$$-1$$

$$\text{at } x = -2$$

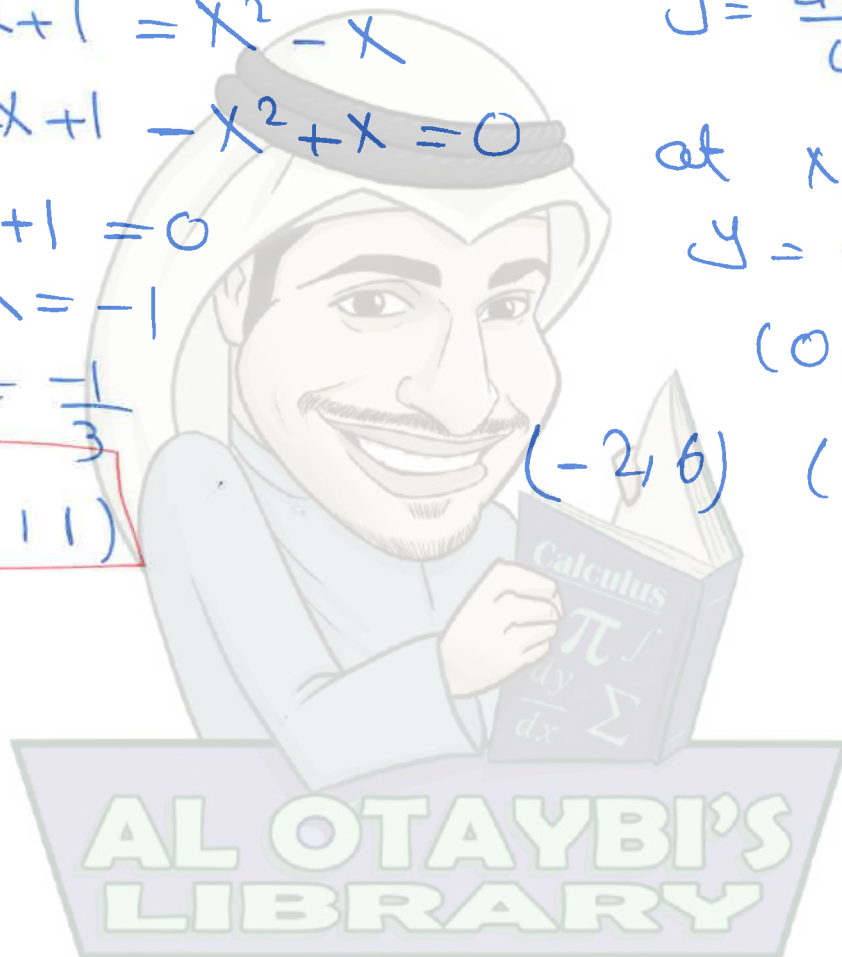
$$y = \frac{4+2}{(-1)^2} = \frac{6}{1} = 6$$

$$\text{at } x = 0$$

$$y = \frac{0}{(1)^2} = 0$$

$$(0, 0)$$

$$(-2, 6) \quad (0, 0)$$



$$Q_F = \{R \mid \varepsilon - 1\}$$

$$H.A \rightarrow y=1$$

$$V.A \rightarrow x=-1$$

$$\oplus +\infty$$

$$\ominus +\infty$$

$$\left(\frac{1}{3}, -\frac{1}{8}\right) \text{ L.min}$$

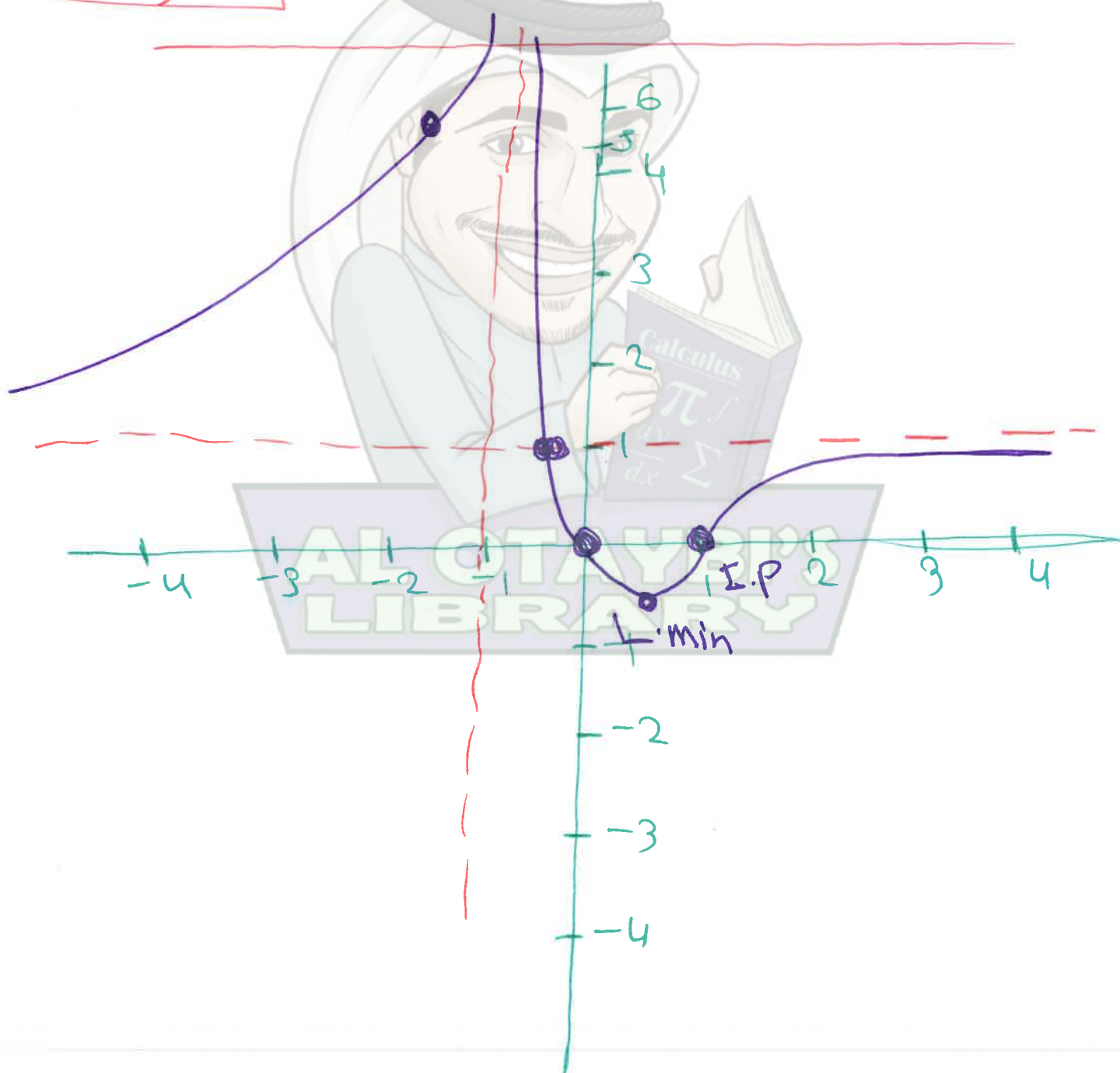
$$(1,0) \text{ I.P}$$

$$\cup (-\infty, -1) \quad (-1, 1) \quad \cap (1, \infty)$$

$$\boxed{\left(-\frac{1}{3}, 1\right)}$$

$$(-2, 6)$$

$$(0, 0)$$



Kuwait University
Department of Mathematics

Calculus I (Math 101)
Second Exam

December 6, 2014
Duration 90 minutes

Answer all questions. Calculators and mobile phones are not allowed.

Q1. [5 pts.] Evaluate the following limit, if it exists:

$$\lim_{x \rightarrow 0} \left(\frac{\sin(3x^2)}{7x^3 - 3x^2} \right).$$

Q2. [8+7=15 pts.] Find $\frac{dy}{dx}$ if:

(a) $e^{xy} + 2y - 3x = \sin(y)$

(b) $y = x^{\sin(x)}$.

Q3. [10 pts.] Oil spilled from a tanker spreads in a circle whose area increases at a constant rate of $9 \text{ km}^2/\text{hr}$. How fast is the radius of the spill increasing when the area is 16 km^2 ?

Q4. [10 pts.] Use linear approximation to estimate: $\ln(1.013)$.

Q5. [10 pts.] Use the Mean Value Theorem to show that for $x > 0$,

$$\tan^{-1}(x) < x.$$

Q6. [50 pts.] Let $f(x) = \frac{x^2 + 1}{(x^2 - 4)^2}$.

a) Study the symmetry of the curve $y = f(x)$ and find the x -intercepts and y -intercepts, if any.

b) Find the vertical and horizontal asymptotes for the graph of f , if any.

c) Given that $f'(x) = \frac{-2x(x^2 + 6)}{(x^2 - 4)^3}$.

i) Find the intervals on which f is increasing and the intervals on which f is decreasing.

ii) Find the local maximum and minimum values of f , if any.

d) Given that $f''(x) = \frac{6(x^4 + 14x^2 + 8)}{(x^2 - 4)^4}$.

i) Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward.

ii) Find the points of inflection, if any.

e) Sketch the graph of f .

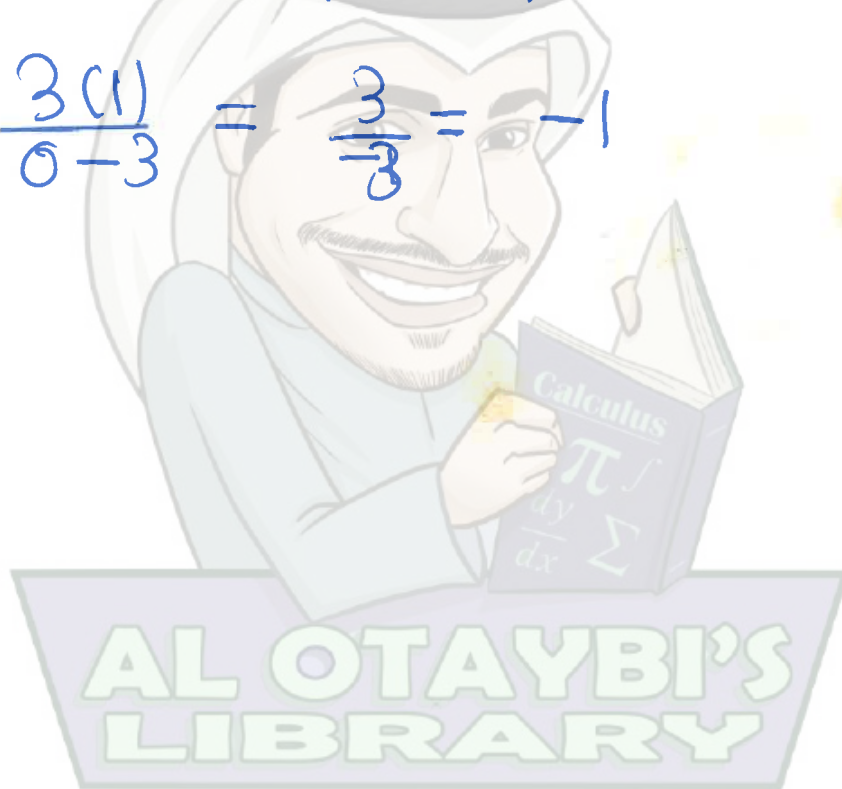
Q#1

$$\lim_{x \rightarrow 0} \left(\frac{\sin(3x^2)}{7x^3 - 3x^2} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{3x^2 \frac{\sin(3x^2)}{3x^2}}{7x^3 - 3x^2} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{\cancel{3x^2} \frac{\sin(3x^2)}{\cancel{3x^2}}}{\cancel{x^2} (7x - 3)} \right)$$

$$= \frac{3(1)}{0-3} = \frac{3}{-3} = -1$$



Q#2 Find $\frac{dy}{dx}$

① $e^{xy} + 2y - 3x = \sin y$

Diff w.r.t. x

$$e^{xy} [y + x y'] + 2y' - 3 = \cos y y'$$

$$e^{xy} y + e^{xy} x y' + 2y' - 3 = \cos y y'$$

$$e^{xy} y - 3 = \cos y y' - 2y' - e^{xy} x y'$$

$$y' [\cos y - 2 - e^{xy} x] = e^{xy} y - 3$$

$$y' = \frac{e^{xy} y - 3}{\cos y - 2 - e^{xy} x}$$

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Find $\frac{dy}{dx}$

(b) $y = x^{\sin x}$

$\ln y = \ln x^{\sin x}$

$\ln y = \sin x \ln x$

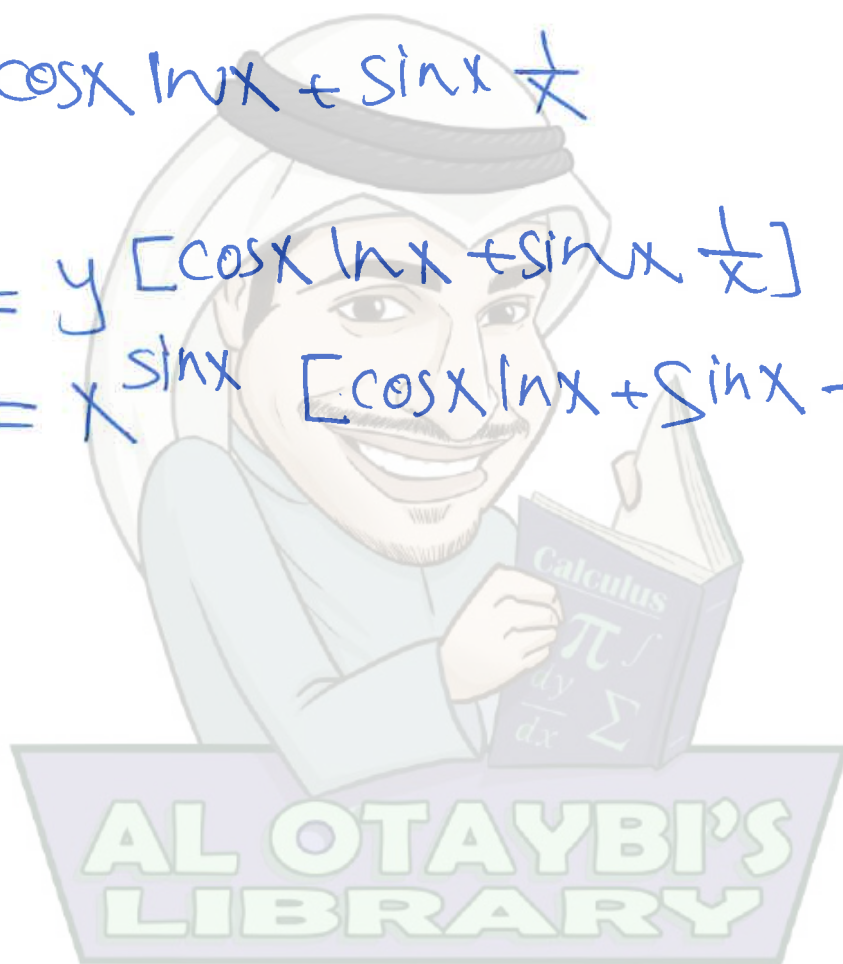
معطى

x^x

المطلوب

$\frac{y'}{y} = \cos x \ln x + \sin x \cdot \frac{1}{x}$

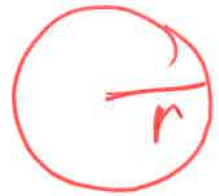
* $y' = y [\cos x \ln x + \sin x \cdot \frac{1}{x}]$
 $= x^{\sin x} [\cos x \ln x + \sin x \cdot \frac{1}{x}]$



Q#3

$$\frac{dA}{dt} = 9 \text{ km}^2/\text{hr}$$

$$\frac{dr}{dt} ?? \rightarrow A = 16 \text{ km}^2$$



$$A = \pi r^2$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$9 = 2\pi \frac{4}{\sqrt{\pi}} \frac{dr}{dt}$$

$$9 = 8 \sqrt{\pi} \frac{1}{\sqrt{\pi}} \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{9}{8\sqrt{\pi}} \text{ km/hr}$$

$$A = \pi r^2$$

$$16 = \pi r^2$$

$$r^2 = \frac{16}{\pi}$$

$$r = \sqrt{\frac{16}{\pi}}$$

$$r = \frac{4}{\sqrt{\pi}}$$



Q#4

$\ln(1.013)$

$$f(x) = \ln x$$

$$x_0 = 1$$

$$x_1 = 1.013$$

$$\rightarrow \Delta x = 1.013 - 1 = 0.013$$

$$f(x_0) = \ln 1 = 0$$

$$f'(x) = \frac{1}{x}$$

$$f'(x_0) = \frac{1}{1} = 1$$

$$\begin{aligned} f(x_1) &\approx f(x_0) + f'(x_0) \Delta x \\ &\approx 0 + 1 \cdot (0.013) \\ &\approx 0.013 \end{aligned}$$

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Q#5

$$\tan^{-1} x \leq x$$

$$x > 0$$

$$f(x) = \tan^{-1} x \quad [0, \infty)$$

① Domain of $f(x) = \mathbb{R}$ [inverse of trigonometric fun.]
∴ f is continuous on $[0, \infty)$

② $f'(x) = \frac{1}{1+x^2}$ [Alg. function]

f is differentiable on $\mathbb{R}$
∴ f is differentiable on $(0, \infty)$

From ① ② ∴ M.V.T is satisfied
then there exist at least one $c \in (0, \infty)$

s.t. $f'(c) = \frac{f(b) - f(a)}{b-a}$

$$\frac{1}{1+c^2} = \frac{f(x) - f(0)}{x-0}$$

$$\frac{1}{1+c^2} = \frac{\tan^{-1} x - \cancel{\tan^{-1} 0}}{x - \cancel{0}}$$

$$\frac{1}{1+c^2} = \frac{\tan^{-1} x}{x}$$

$$0 < C < x$$

$$0 < C^2 < x^2$$

(+1)

$$1 < 1+C^2 < 1+x^2$$

$$1 > \frac{1}{1+C^2} > \frac{1}{1+x^2}$$



$$1 > \frac{\tan^{-1}x}{x} > \frac{1}{1+x^2}$$

$$1 > \frac{\tan^{-1}x}{x}$$

$$x > \tan^{-1}x$$

#

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$$f(x) = \frac{x^2+1}{(x^2-4)^2}$$

Domain of $f(x)$: $x^2-4=0$
 $x^2=4 \rightarrow x = \pm 2$
 $\mathbb{R} \setminus \{\pm 2\}$

HA

$$\lim_{x \rightarrow \pm \infty} \frac{x^2(1+\frac{1}{x^2})}{[x^2(1-\frac{4}{x^2})]^2}$$

$$\lim_{x \rightarrow \pm \infty} \frac{\cancel{x^2}(1+\frac{1}{x^2})}{x^4(1-\frac{4}{x^2})^2}$$

$$\lim_{x \rightarrow \pm \infty} \frac{1+\frac{1}{x^2}}{x^2(1-\frac{4}{x^2})^2} = \frac{1+0}{\infty} = 0$$

$\Rightarrow \infty$ $y=0$ is HA.

V.A $x=2$

$$\lim_{x \rightarrow 2^+} \frac{x^2+1}{(x^2-4)^2} = +\infty$$

$$\lim_{x \rightarrow 2^-} \frac{x^2+1}{(x^2-4)^2} = +\infty$$

$\Rightarrow \infty$ $x=2$ is V.A.

$x=-2$

$$\lim_{x \rightarrow -2^+} \frac{x^2+1}{(x^2-4)^2} = +\infty$$

$$\lim_{x \rightarrow -2^-} \frac{x^2+1}{(x^2-4)^2} = +\infty$$

$\Rightarrow \infty$ $x=-2$ is V.A.

$$f'(x) = \frac{-2x(x^2+6)}{(x^2-4)^3}$$

$\downarrow$
 $=0$

$$-2x(x^2+6)=0$$

$$-2x=0$$

$$x=0$$

$\in \text{Dom}$

$$x^2+6=0$$

$$x^2=-6$$

Nohe

$\rightarrow$

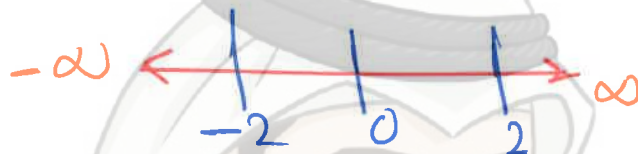
DNE

$$x^2-4=0$$

$$x^2=4$$

$$x=\pm 2$$

$\notin \text{Domain}$



$f'(x)$	$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$
$\frac{-2x(x^2+6)}{(x^2-4)^3}$	+	+	-	-
	+	-	-	+

$\oplus$

$\rightarrow$

$\ominus$

$\rightarrow$

$\oplus$

$\rightarrow$

$\ominus$

$\rightarrow$

$\therefore f(x)$ is increasing on $(-\infty, -2)$ $(0, 2)$

$f(x)$ is decreasing on $(-2, 0)$ $(2, \infty)$

$$\text{at } x=0 \rightarrow y = \frac{0+1}{(0-4)^2} = \frac{1}{16}$$

$\Rightarrow (0, \frac{1}{16})$ L. min

$$f''(x) = \frac{6(x^4 + 14x^2 + 8)}{(x^2 - 4)^4}$$

$$= 0 \swarrow$$

$\searrow$ DNE

$$\div 6 \quad 6(x^4 + 14x^2 + 8) = 0$$

$$x^4 + 14x^2 + 8 = 0$$

$$x^4 + 14x^2 = -8$$

None

$$x^2 - 4 = 0$$

$$x^2 = 4$$

$$x = \pm 2$$

$\notin$ Domain



$f''(x)$	$(-\infty, -2)$	$(-2, 2)$	$(2, \infty)$
$\frac{6(x^4 + 14x^2 + 8)}{(x^2 - 4)^4}$	+	+	+
	$\oplus$	$\oplus$	$\oplus$

$\therefore f(x)$ is concave up on $(-\infty, -2)$ $(-2, 2)$ $(2, \infty)$

No interval of concave down

No inflection points

key points

intersection with
HA

-3 1 3
 -2 2

$$0 = \frac{x^2+1}{(x^2-4)^2}$$

$$0 = x^2+1$$
$$x^2 = -1$$

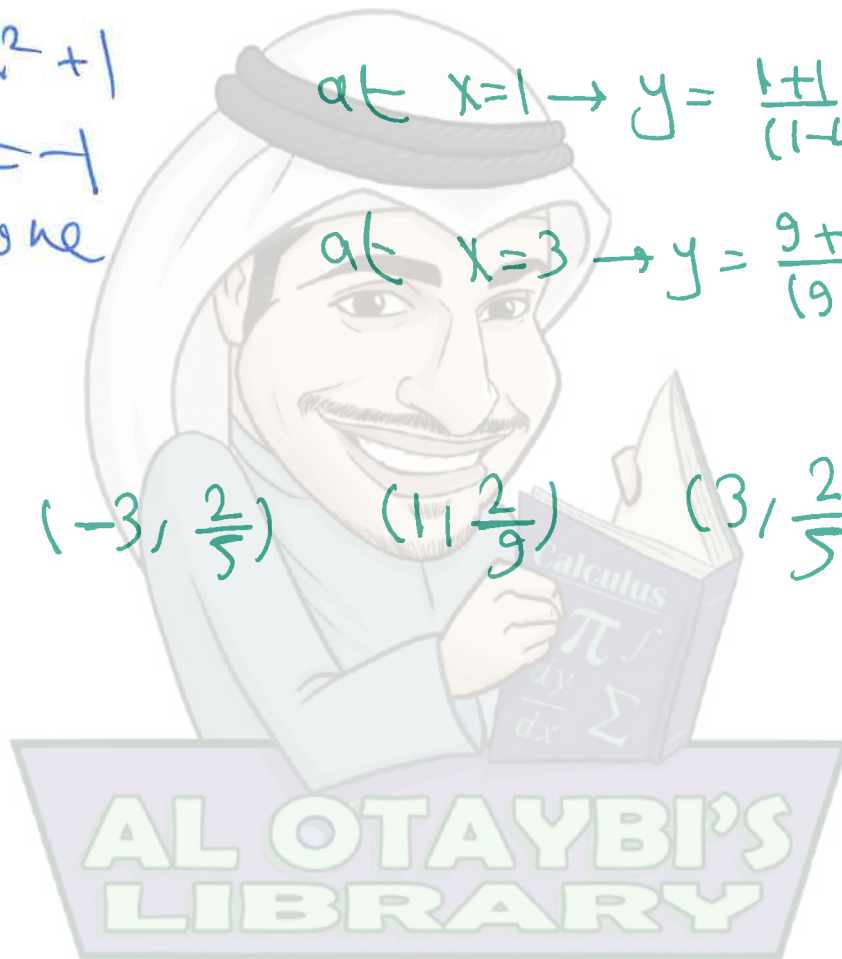
None

$$\text{at } x=-3 \rightarrow y = \frac{9+1}{(9-4)^2} = \frac{10}{5^2} = \frac{10}{25} = \frac{2}{5}$$

$$\text{at } x=1 \rightarrow y = \frac{1+1}{(1-4)^2} = \frac{2}{(-3)^2} = \frac{2}{9}$$

$$\text{at } x=3 \rightarrow y = \frac{9+1}{(9-4)^2} = \frac{10}{5^2} = \frac{10}{25} = \frac{2}{5}$$

$\Rightarrow (-3, \frac{2}{5}) \quad (1, \frac{2}{9}) \quad (3, \frac{2}{5})$



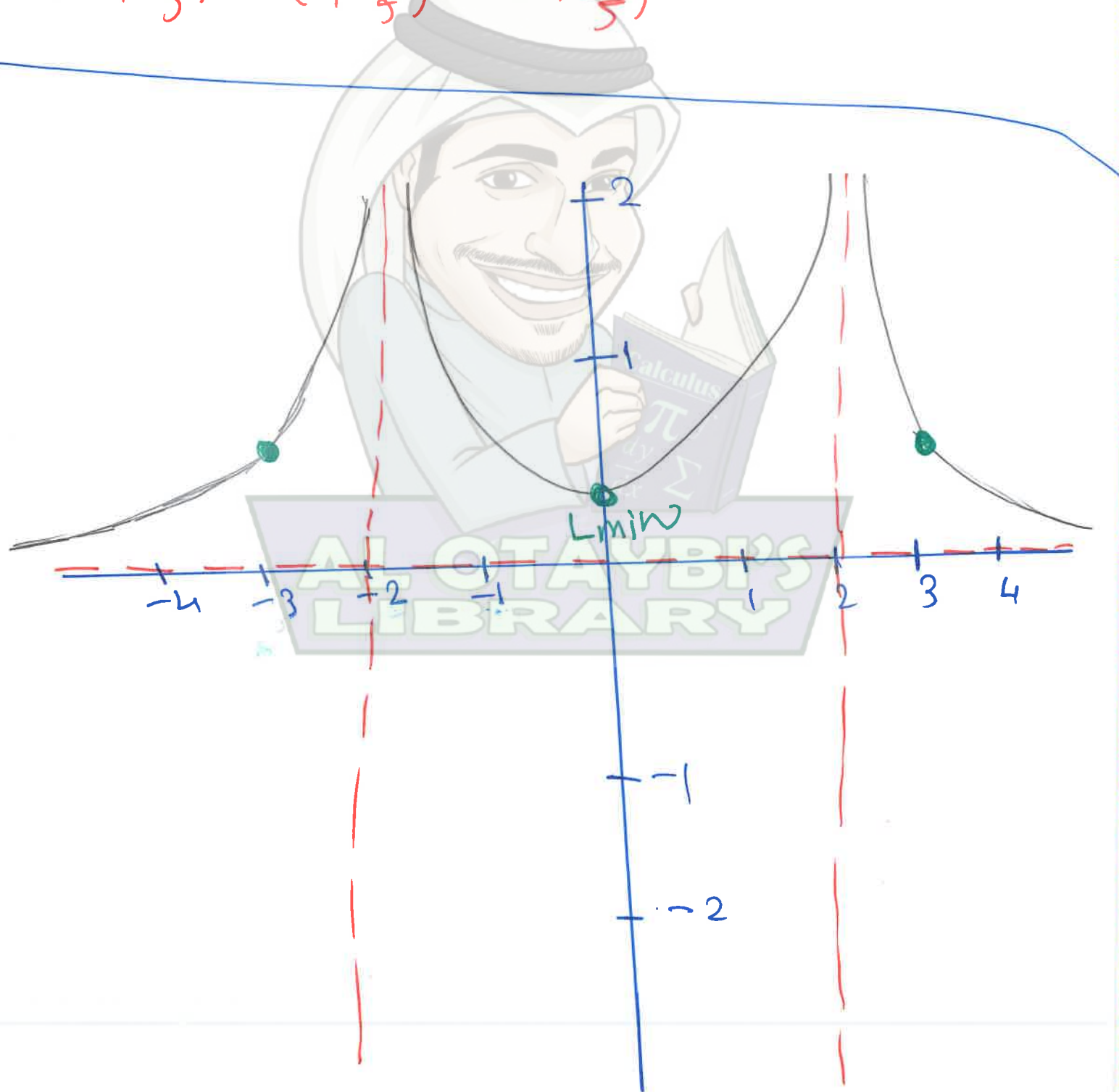
HA $\rightarrow y = 0$

V.A $\rightarrow x = 2 \rightarrow \begin{cases} \oplus +\infty \\ \ominus +\infty \end{cases}$
 $\rightarrow x = -2 \rightarrow \begin{cases} \oplus +\infty \\ \ominus +\infty \end{cases}$

$(0, \frac{1}{16})$ L.min

$\cup (-\infty, -2) \quad (-2, 2) \quad (2, \infty)$

$(-3, \frac{2}{5}) \quad (1, \frac{2}{5}) \quad (3, \frac{2}{5})$



$$f(x) = \frac{x^2 + 1}{(x^2 - 4)^2}$$

$$a) f(-x) = \frac{(-x)^2 + 1}{((-x)^2 - 4)^2} = \frac{x^2 + 1}{(x^2 - 4)^2} = f(x)$$

$\therefore f(-x) = f(x) \rightarrow \therefore f$ is symmetric about y-axis.

* intersection *

y-axis x-axis

$$y = \frac{0+1}{(0-4)^2} = \frac{1}{16}$$

$$\Rightarrow (0, \frac{1}{16})$$

$$\textcircled{0} = \frac{x^2 + 1}{(x^2 - 4)^2}$$

$$\textcircled{0} = x^2 + 1$$

$$x^2 = -1$$

None



Kuwait University
Department of Mathematics

Calculus I (Math 101)
Second Exam

19 April 2014
Duration 90 minutes

Answer all questions. Calculators and mobile phones are not allowed.

Q1. [10 pts.] Let $f(x) = (2 + \sin(1 - x^2))^4$. Find $f'(1)$.

Q2. [10 pts.] Find an equation of the tangent line at the point $(0, 3)$ to the curve

$$y^3 + x^2y + x^2 - 3y^2 = 0.$$

Q3. [10 pts.] Let $f(x) = \sqrt{1 + 4x}$. Use linear approximation to estimate $f(2.03)$.

Q4. [10 pts.] Show that $f(x) = \sqrt{16 - x^2}$ satisfies the conditions of Rolle's Theorem on the interval $[-2, 2]$. Then find all numbers c that satisfy the conclusion of Rolle's Theorem.

Q5. [10 pts.] The length of a rectangle is increasing at a rate of 10 cm/s and its width is decreasing at a rate of 4 cm/s. When the length is 40 cm and the width is 15 cm, how fast is the area of the rectangle changing?

Q6. [50 pts.] Let $f(x) = \frac{4 - x^2}{x^2 - 9}$.

- a) Study the symmetry of the curve and find the x -intercepts and y -intercepts, if any.
- b) Find the vertical and horizontal asymptotes for the graph of f , if any.

c) Given that $f'(x) = \frac{10x}{(x^2 - 9)^2}$.

- i) Find the intervals on which f is increasing and the intervals on which f is decreasing.
- ii) Find the local maximum and minimum values of f , if any.

d) Given that $f''(x) = \frac{-30(x^2 + 3)}{(x^2 - 9)^3}$.

- i) Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward.
 - ii) Find the points of inflection, if any.
- e) Sketch the graph of f .

Q #1 Let $f(x) = (2 + \sin(1 - x^2))^4$
Find $f'(1)$.

$$f'(x) = 4 [2 + \sin(1 - x^2)]^3 \cos(1 - x^2) (-2x)$$

at $x=1$

$$\begin{aligned} f'(1) &= 4 [2 + 0]^3 (1) (-2) \\ &= 4 (8) (-2) = -64 \end{aligned}$$

Q #2 Find the equation of tangent line at $(0, 3)$

$$y^3 + x^2 y + x^2 - 3y^2 = 0$$

Diff. w.r.t. x

اشتقاق y^3 + اشتقاق $x^2 y$ + اشتقاق x^2 - اشتقاق $3y^2$

$$3y^2 y' + 2xy + x^2 y' + 2x - 6y y' = 0$$

at $(0, 3)$

$$3(9) y' + 0 + 0 + 0 - 6(3) y' = 0$$

$$27 y' - 18 y' = 0$$

$$9 y' = 0 \rightarrow y' = 0$$

∴ the equation of tangent line: $y - y_0 = m(x - x_0)$

$$y - 3 = 0(x - 0)$$

$$y - 3 = 0$$

Q#3

Let $f(x) = \sqrt{1+4x}$
estimate $f(2.03)$

$$f(x) = \sqrt{1+4x}$$

$$x_0 = 2$$

$$x_1 = 2.03$$

$$\longrightarrow \Delta x = 2.03 - 2 = 0.03$$

① $f(x_0) = \sqrt{1+4(2)} = \sqrt{1+8} = \sqrt{9} = 3$

② $f'(x) = \frac{4}{2\sqrt{1+4x}} = \frac{2}{\sqrt{1+4x}}$

③ $f'(x_0) = \frac{2}{\sqrt{1+8}} = \frac{2}{\sqrt{9}} = \frac{2}{3}$

$$f(x_1) \approx f(x_0) + f'(x_0) \Delta x$$

$$\approx 3 + \frac{2}{3} (0.03)$$

$$\approx 3 + \frac{2}{3} \frac{3}{100}$$

$$\approx 3 + \frac{1}{50}$$

$$\approx \frac{150}{50} + \frac{1}{50}$$

$$\approx \frac{151}{50}$$

Q#1

$$f(x) = \sqrt{16-x^2}$$

$$[-2, 2]$$

① Domain of $f(x)$: $16 - x^2 \geq 0$

$$16 \geq x^2$$

$$[-4, 4]$$

$\therefore D_f = [-4, 4]$

$\therefore f$ is continuous on $[-2, 2]$ [Alg. function]

② $f'(x) = \frac{-2x}{2\sqrt{16-x^2}}$ [Alg. function]

f is differentiable on $(-4, 4)$

$\therefore f$ is differentiable on $(-2, 2)$

③ $f(-2) = \sqrt{16-4} = \sqrt{12}$
 $f(2) = \sqrt{16-4} = \sqrt{12} \rightarrow \therefore f(-2) = f(2)$

From ①②③ $\rightarrow \therefore$ Rolle's Thm is satisfied
then there is a number $c \in (-2, 2)$

s.t. $f'(c) = 0$

$$\frac{-c}{\sqrt{16-c^2}} = 0$$

$$c = 0$$

Q#5

$$\frac{dL}{dt} = 10 \text{ cm/s}$$

$$\frac{dw}{dt} = -4 \text{ cm/s}$$

$$\frac{dA}{dt} \rightarrow \begin{aligned} L &= 40 \text{ cm} \\ w &= 15 \text{ cm} \end{aligned}$$



$$\begin{aligned} A &= L \times w \\ C &= 2L + 2w \end{aligned}$$

$$A = L \times w$$

Diff. w.r.t.t

$$\frac{dA}{dt} = \frac{dL}{dt} w + L \frac{dw}{dt}$$

$$= 10(15) + 40(-4)$$

$$= 150 - 160$$

$$= -10 \text{ cm}^2/\text{s}$$

Q#6

$$f(x) = \frac{4-x^2}{x^2-9}$$

① Domain of $f(x)$:

$$x^2 - 9 = 0 \rightarrow x^2 = 9 \rightarrow x = \pm 3$$

$$\Rightarrow \underline{D_f} = \mathbb{R} \setminus \{\pm 3\}$$

② HA

$$\lim_{x \rightarrow \pm \infty} \frac{4-x^2}{x^2-9}$$

$$\lim_{x \rightarrow \pm \infty} \frac{x^2(\frac{4}{x^2}-1)}{x^2(1-\frac{9}{x^2})} = \frac{0-1}{1-0} = -1$$

$\therefore y = -1$ is HA

③ V.A $x=3$

$$\lim_{x \rightarrow 3^+} \frac{4-x^2}{(x-3)(x+3)} = \frac{-5}{+0} = +\infty$$

$x \rightarrow 3^+ \rightarrow 3 + 0.1 = 3.1$

$$\lim_{x \rightarrow 3^-} \frac{4-x^2}{(x-3)(x+3)} = \frac{-5}{-0} = +\infty$$

$x \rightarrow 3^- \rightarrow 3 - 0.1 = 2.9$

$\therefore x=3$ is V.A

$$x=-3$$

$$\lim_{x \rightarrow -3^+} \frac{4-x^2}{(x-3)(x+3)} = \frac{-5}{-0} = +\infty$$

$-3 + 0.1 = -2.9$

$$\lim_{x \rightarrow -3^-} \frac{4-x^2}{(x-3)(x+3)} = \frac{-5}{+0} = -\infty$$

$-3 - 0.1 = -3.1$

$\therefore x=-3$ is V.A

$$f'(x) = \frac{10x}{(x^2-9)^2}$$

$$= 0$$

$$10x = 0$$

$$x = 0$$

$\in \text{Dom.}$

undefined (DNE)

$$x^2 - 9 = 0$$

$$x^2 = 9$$

$$x = \pm 3$$

$\notin \text{Domain}$



	$(-\infty, -3)$	$(-3, 0)$	$(0, 3)$	$(3, \infty)$
$\frac{10x}{(x^2-9)^2}$	-	-	+	+
	+	+	+	+

$\ominus$

$\ominus$

$\oplus$

$\oplus$

$\therefore$ interval of increasing = $(0, 3) (3, \infty)$
interval of decreasing = $(-3, 0) (-\infty, -3)$

at $x = 0 \rightarrow \text{L.min}$

$$y = \frac{4-0}{0-9} = \frac{4}{-9} = -\frac{4}{9}$$

$\Rightarrow (0, -\frac{4}{9}) \text{ L.min.}$

$$f''(x) = \frac{-30(x^2+3)}{(x^2-9)^3}$$

$$= 0$$

$$-30(x^2+3) = 0$$

$$\div (-30)$$

$$x^2 + 3 = 0$$

$$x^2 = -3$$

None

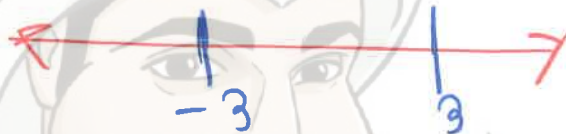
undefined (DNE)

$$x^2 - 9 = 0$$

$$x^2 = 9$$

$$x = \pm 3$$

$\notin$ Domain



$f''(x)$	$(-\infty, -3)$	$(-3, 3)$	$(3, \infty)$
$\frac{-30(x^2+3)}{(x^2-9)^3}$	-	-	-
	+	-	+
	$\ominus$	$\oplus$	$\ominus$
	$\cap$	$\cup$	$\cap$

$\therefore f(x)$ is concave up on $(-3, 3)$

$f(x)$ is concave down on $(-\infty, -3) (3, \infty)$

No inflection point

intersection point
with HA

$$\begin{array}{ccc} -4 & | & 1 \\ & | & \\ -3 & | & 3 \end{array}$$

$$-1 = \frac{4-x^2}{x^2-9}$$

$$-(x^2-9) = 4-x^2$$

$$-x^2 + 9 = 4 - x^2$$

$$9 = 4$$

one

at $x = -4$

$$y = \frac{4-16}{16-9} = -\frac{12}{7} = -\frac{12}{7}$$

at $x = 4$

$$y = \frac{4-16}{16-9} = -\frac{12}{7}$$

$$\Rightarrow (-4, -\frac{12}{7})$$

$$(4, -\frac{12}{7})$$

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* Intercept *

x-intercept

y-intercept

$$0 = \frac{4 - x^2}{x^2 - 9}$$

$$0 = 4 - x^2$$

$$x^2 = 4$$

$$x = 2$$

$$(2, 0)$$

$$x = -2$$

$$(-2, 0)$$

$$y = \frac{4 - 0}{0 - 9}$$

$$y = \frac{4}{-9}$$

$$y = -\frac{4}{9}$$

$$(0, -\frac{4}{9})$$

* Symmetry *

$$f(-x) = \frac{4 - (-x)^2}{(-x)^2 - 9} = \frac{4 - x^2}{x^2 - 9}$$

$$= f(x)$$

∴ $f(x)$ is Even.

∴ $f(x)$ is symmetry about y-axis

$$D_f : |R| \{ \pm 3 \}$$

$$HA : y = -1$$

$$V.A \rightarrow x = 3 \begin{cases} \oplus -\infty \\ \ominus +\infty \end{cases}$$

$$x = -3 \begin{cases} \oplus +\infty \\ \ominus -\infty \end{cases}$$

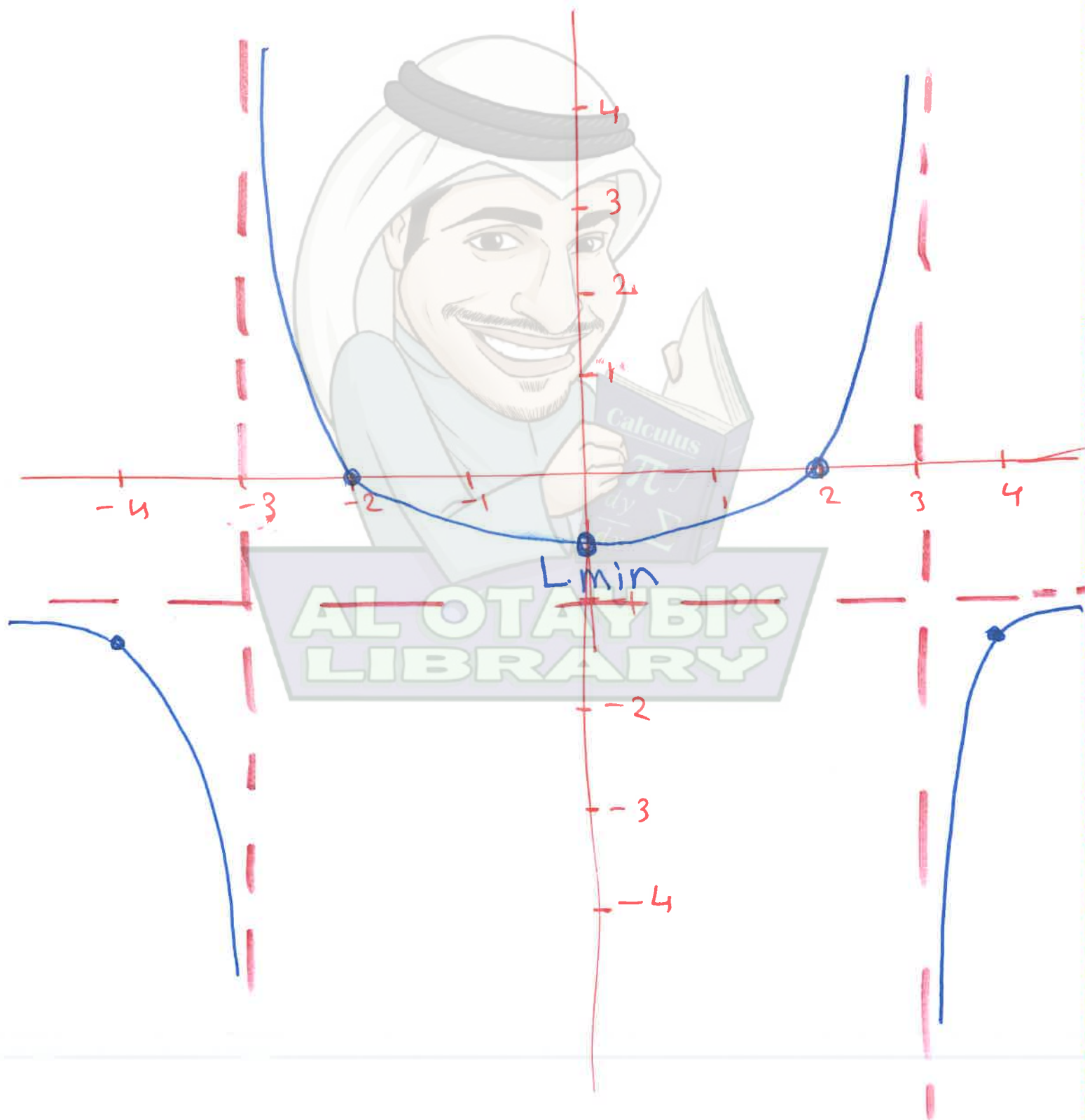
$$(0, -\frac{4}{9}) L.min$$

$$U (-3, 3)$$

$$\cap (-\infty, -3) (3, \infty)$$

$$(-4, -\frac{12}{7})$$

$$(4, -\frac{12}{7})$$



Kuwait University
Department of Mathematics

Calculus I (Math 101)
Second Exam

17 December 2016
Duration 90 minutes

Answer all questions. Calculators and mobile phones are not allowed.

1. [10 pts.] Evaluate: $\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2 + x - 6}$.
 2. [10 pts.] Find an equation of the tangent line to the curve $y = x^{\ln(x)}$ at the point $(1, 1)$.
 3. [10 pts.] A cylindrical tank of diameter 6 m is being filled with water at a rate of $2 \text{ m}^3/\text{hr}$, how fast is the water level rising?
 4. [10 pts.] Let $f(x) = e^x \cos x$. Use linear approximation to estimate $f(0.1)$.
 5. [10 pts.] Find the absolute maximum and minimum values of the function $f(x) = 3x - 5x^{3/5}$ on the interval $[-1, 32]$.
 6. [5+5=10 pts.] Let $f(x) = \ln(x^2 + 1)$.
 - (a) Verify that f satisfies the conditions of Rolle's Theorem on $[-1, 1]$.
 - (b) Find all $c \in (-1, 1)$ that satisfy the conclusion of Rolle's Theorem.
 7. [40 pts.] Let $f(x) = 1 + e^{-x^2}$.
 - (a) Find the horizontal asymptotes of the graph of f , if any.
 - (b) Given that $f'(x) = -2xe^{-x^2}$.
 - i) Find the intervals on which f is increasing and the intervals on which f is decreasing, if any.
 - ii) Find the local maximum and minimum values of f , if any.
 - (c) Given that $f''(x) = 2(2x^2 - 1)e^{-x^2}$.
 - i) Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward, if any.
 - ii) Find the points of inflection, if any.
 - (d) Sketch the graph of f .
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بال
class

$$1) \lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2 + x - 6}$$

0/0

$$\frac{x-2}{2-x}$$

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)(x+3)}$$

$$\lim_{x \rightarrow 2} \frac{\cancel{(x-2)} \sin(x-2)}{\cancel{(x-2)} (x+3)}$$

$$= \frac{1}{5}$$

Sol #2

$$\lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)(x+3)}$$

$$= (1) \frac{1}{5} = \frac{1}{5}$$

2) Find the equation of the tangent line to $y = x^{\ln x}$ at (1,1)

$$\ln y = \ln x^{\ln x}$$

$$\ln y = \ln x \ln x$$

$$\ln y = (\ln x)^2$$

Diff. w.r.t. x

$$\frac{y'}{y} = 2(\ln x) \frac{1}{x}$$

$$*y \quad y' = y \left[2(\ln x) \frac{1}{x} \right]$$

at (1,1)

$$y' = (1) [0] = 0$$

∴ Equation of tangent line

$$y - y_0 = m(x - x_0)$$

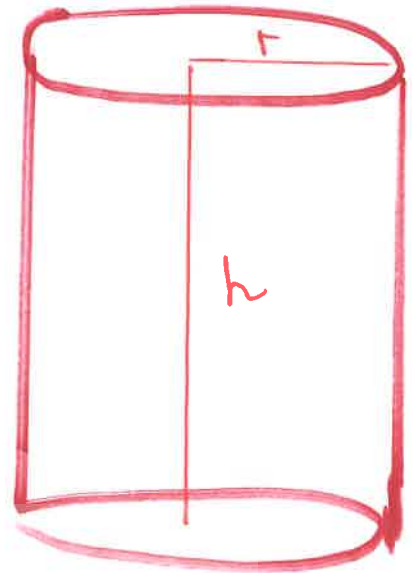
$$y - 1 = 0(x - 1)$$

$$y - 1 = 0$$

3) $d = 6\text{m} \rightarrow r = 3\text{m}$

$$\frac{dV}{dt} = 2 \text{ m}^3/\text{hr}$$

$$\frac{dh}{dt} ??$$



$$V = \pi r^2 h$$

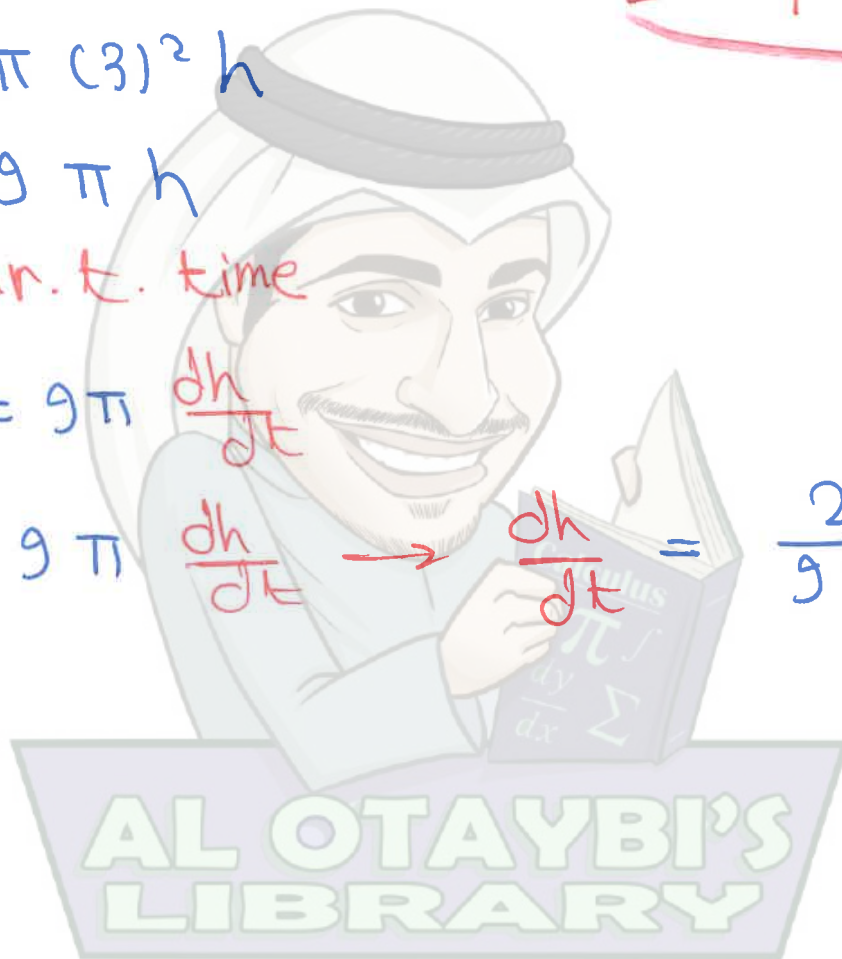
$$V = \pi (3)^2 h$$

$$V = 9\pi h$$

Diff. w.r.t. time

$$\frac{dV}{dt} = 9\pi \frac{dh}{dt}$$

$$2 = 9\pi \frac{dh}{dt} \rightarrow \frac{dh}{dt} = \frac{2}{9\pi} \text{ m/hr}$$



4) $f(x) = e^x \cos x$ estimate $f(0.1)$

$$f(x) = e^x \cos x$$

$$x_0 = 0$$

$$x_1 = 0.1 \rightarrow \Delta x = x_1 - x_0 = 0.1 - 0 = 0.1$$

① $f(x_0) = e^0 \cos 0 = (1)(1) = 1$

② $f'(x) = e^x \cos x + e^x (-\sin x)$

③ $f'(x_0) = e^0 \cos 0 + 0 = (1)(1) = 1$

$$\begin{aligned} f(x_1) &\approx f(x_0) + f'(x_0) \Delta x \\ &\approx 1 + 1 \cdot (0.1) \end{aligned}$$

$$\begin{aligned} &\approx 1 + 0.1 \\ &\approx 1.1 \end{aligned}$$

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⑤ $f(x) = 3x - 5x^{\frac{3}{5}}$

Find abs. max
abs. min
on $[-1, 32]$

Domain of $f(x) = \mathbb{R}$

$$f'(x) = 3 - \cancel{5} \cdot \frac{3}{\cancel{5}} x^{-\frac{2}{5}}$$

$$= 3 - \frac{3}{x^{\frac{2}{5}}}$$

$$= \frac{3x^{\frac{2}{5}}}{x^{\frac{2}{5}}} - \frac{3}{x^{\frac{2}{5}}}$$

$$= \frac{3x^{\frac{2}{5}} - 3}{x^{\frac{2}{5}}}$$

$= 0$

$$3x^{\frac{2}{5}} - 3 = 0$$

$$3[x^{\frac{2}{5}} - 1] = 0$$

$$\div 3 \quad x^{\frac{2}{5}} - 1 = 0$$

$$x^{\frac{2}{5}} = 1$$

undefined (DNE)

$$x^{\frac{2}{5}} = 0$$

$$x = 0$$

$\in [-1, 32]$

$$x = 1$$

$\in [-1, 32]$

$$f(0) = 0 - 0 = 0$$

$$f(1) = 3 - 5 = -2$$

$$f(-1) = -3 - 5(-1)^{\frac{3}{5}} = -3 + 5 = -2$$

$$\begin{aligned} f(32) &= 3(32) - 5(32)^{\frac{3}{5}} \\ &= 96 - 5(8) = 96 - 40 = 56 \end{aligned}$$

$\therefore \text{abs. max} = 56$

$\therefore \text{abs. min} = -2$

$$6) f(x) = \ln(x^2 + 1) \quad [-1, 1]$$

$$1) \text{ Domain of } f(x) : \begin{matrix} x^2 + 1 \geq 0 \\ x^2 \geq -1 \end{matrix}$$

$$\Rightarrow \text{Domain of } f(x) = \mathbb{R}$$

$\therefore f(x)$ is continuous on $[-1, 1]$

$$2) f'(x) = \frac{2x}{x^2 + 1}$$

f is differentiable on :

$$\begin{matrix} x^2 + 1 = 0 \\ x^2 = -1 \end{matrix}$$

$\Rightarrow \therefore f$ is differentiable on $\mathbb{R}$

$\therefore f$ is differentiable on $(-1, 1)$

$$3) f(-1) = \ln(1+1) = \ln 2 \rightarrow \therefore f(-1) = f(1)$$

$$f(1) = \ln(1+1) = \ln 2$$

From ① ② ③ $\rightarrow \therefore$ Rolle's Thm is satisfied

then there exist at least one $c \in (-1, 1)$

s.t

$$f'(c) = 0$$

$$\frac{2c}{c^2 + 1} = 0$$

$$* (c^2 + 1)$$

$$2c = 0$$

$$c = 0$$

$$\in (-1, 1)$$

